FORENSIC AUDIT · V3 COMPARISON · DEVELOPMENT BASELINE

PAVEMENT-PINN v3

Consolidated Forensic Audit, Reference-Code Comparison & Development Baseline

Revalidation of AUDIT-001, traceability to the v3 specification, independent assessment of the reference implementation, and a gated roadmap for the next development phases.

NOT v3-COMPLIANT — 19 OPEN P0 ITEMS — PHASE 0 RECOMMENDED

AS-IS IMPLEMENTATION · V3 SPECIFICATION · REFERENCE ZIP · VERIFIED FINDING · PROPOSED TARGET STATE

Document ID	PAVEMENT-PINN-V3-BASELINE-001
Version / date	1.0 · 2026-10-15
Repository commit	`4a9b4fc49ec6e1d583648b2eaaa26293577add2e`
Branch	`feature/m1-report-ownership` (working tree with another session's uncommitted changes, preserved)
Status	Issued for team review — project status against v3: NOT v3-COMPLIANT (19 open P0 items)
Specification	PAVEMENT-PINN v3 Dokumen Induk v3.0 (15 Oct 2026, 65 pp.) and AGENTS.md
Reference code	pavement_pinn_v3_code.zip — inspected in isolation, not executed
Previous audit	PAVEMENT-PINN-AUDIT-001 v1.0 (14 Oct 2026)
Classification	Internal engineering document of the project team; not approved by Intel or any agency

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Document Control

Table 0.1 — Document metadata

Field	Value
Document ID	PAVEMENT-PINN-V3-BASELINE-001
Title	PAVEMENT-PINN v3 — Consolidated Forensic Audit, Reference-Code Comparison & Development Baseline
Version / date	1.0 · 2026-10-15
Repository	`pavement-pinn` (local folder …/`Intel AI Global Impact Festival 2026/Projects/pavement-pinn`)
Branch / commit	`feature/m1-report-ownership` · `4a9b4fc49ec6e1d583648b2eaaa26293577add2e`
Working tree	24 modified, 1 deleted, 10 untracked paths from another session's demo-authentication work (unchanged since AUDIT-001), plus the audit folders. Nothing was staged, committed, stashed, reset or overwritten by this baseline
Inputs	`AGENTS.md` (sha256 `06a7635c…4b535`); *PAVEMENT-PINN v3 Dokumen Induk*, 65 pages, v3.0 dated 15 Oct 2026 (sha256 `1cdc4229…b6ab`); `pavement_pinn_v3_code.zip` (sha256 `664c6a7f…2b88`); AUDIT-001 v1.0 (2026-10-14) and its registers
Method	Read-only repository inspection; full reading of AGENTS.md and the 65-page PDF (text layer); isolated, non-executed inspection of the ZIP; independent numerical checks with the AUDIT-001 layered-elastic solver and JavaScript ports (`evidence/`)
Not done	No Python executed (blocked by machine policy, F-38); no `npm test` / `npm run lint` re-run (code identical to the audited state, see §1.4); no build, deploy, migration, dependency installation or Git state change
Status	Baseline issued for team review. Overall project status against v3: **NOT v3-COMPLIANT** — 19 open P0 items
Classification	Internal engineering document of the project team. Not approved by, and not affiliated with, Intel or any government agency
Companion files	Appendices A–G as separate Markdown registers in this folder; `evidence/`; this PDF and its generated Markdown master

Table 0.2 — Status labels used in this report

Label	Meaning
AS-IS IMPLEMENTATION	What the repository actually contains today, from source inspection
V3 SPECIFICATION	What the v3 Dokumen Induk or AGENTS.md requires; not evidence that it exists
REFERENCE ZIP	Candidate code from the other team's package; not executed here
VERIFIED FINDING	Established by evidence collected or re-checked in this baseline
PROPOSED TARGET STATE	Recommended future state; does not exist
TERBUKTI / TARGET / RENCANA / DILARANG	v3 §1.1 claim legend: proven in this repository / measurable goal not yet measured / not built / prohibited claim

Chapter 0 — Executive Summary

Central question. Given the repository's actual state, the 14 October forensic audit, the v3 specification and the reference team's code, what should be kept, corrected, integrated, redesigned, tested and built next?

Short answer. Keep the workflow backbone; remove every unmeasured number first; then build the physics path in a separate Python service using the reference code only after the team reproduces it and fixes the defects found here. Nothing in the AI or physics stack of v3 exists in this repository yet, and the machine used for this baseline cannot run Python at all, which is now the main schedule risk.

Table 0.3 — Executive summary

Area	Finding	Label
Repository state	Code unchanged since AUDIT-001: no file in `src/`, `server/`, `tests/` is newer than the audit register	VERIFIED FINDING
Previous audit	31 findings re-validated: 22 confirmed open, 7 superseded by v3 but still shipped, 1 partially resolved (F-15: decided in v3, not implemented), 1 blocked. None fixed, none regressed	VERIFIED FINDING
Prohibited claims	Six of the eight v3 §1.5 prohibited claims are still visible in the current build (PINN solver, unmeasured latency/NPU, WIM/FWD validation, "aman/melebihi batas ijin", days-to-pothole, agency names)	VERIFIED FINDING
Structural engine	The current engine is a heuristic formula, not a solver and not a network. Its strains are 15× (εt) and 47× (εv) below layered-elastic theory on the v3 reference case	VERIFIED FINDING
MLET reference case	The v3 reference values (εt 303 με, εv 793 με at 10 t; 274/663 at 8.16 t; exact ×3 similarity) are reproduced to 0.1 % by the audit's independent solver	VERIFIED FINDING
Reference ZIP	11 safe files; core math (AI transfer functions, life index, ESAL example, priority example, knapsack property, number checker examples) re-derived and correct. Five design defects found before any training: missing d/a feature (F-32), physically wrong monotonicity signs (F-33), incomplete validation (F-34), no guardrail code (F-35), Monte Carlo seeding/zero-growth issues (F-36)	VERIFIED FINDING
"DIUJI" status	Reported by the other team; not inherited. No reference file was executed here	REFERENCE ZIP
Surrogate / PINN	No trained surrogate, dataset, weights or metrics exist anywhere. All surrogate accuracy numbers in v3 are "contoh dokumen"	TARGET
Benchmarks	None exist; any latency/NPU figure would be fabricated	RENCANA
Privacy	Stored citizen photos keep EXIF/GPS metadata; no blurring; no deletion job (F-46)	VERIFIED FINDING
Environment	Python is blocked by Windows Application Control (F-38); all Python phases blocked on this machine	VERIFIED FINDING
Recommendation	Start with Phase 0 "Truthful status" (pure TypeScript, needs only approvals); resolve HD-01 (Python runtime) in parallel; do not train or benchmark anything until R1-02 reproduces the reference tests on the team's own Intel machine	PROPOSED TARGET STATE

What must not be claimed today: that the project is v3-compliant; that a PINN or PI-surrogate exists or was trained; that the reference code passed tests in this repository; any latency, NPU, accuracy or energy figure; field (FWD/WIM) validation; that the system is offline-capable.

Chapter 1 — Audit Provenance and Document Authority

1.1 Source hierarchy

Table 1.1 — Sources and their authority

Source	Role in this baseline	Authority
A — Current repository	Object of the audit; the only evidence of what exists	Highest for "what is"
B — AUDIT-001 (2026-10-14) and registers	Historical baseline; every finding re-validated (Appendix C)	Historical; superseded by new evidence
C1 — v3 Dokumen Induk (65 pp.)	Target specification; decisions D-01–D-18	Highest for "what should be" (D-01)
C2 — AGENTS.md	Operational rules for agents; defers to the PDF on conflict	Below the PDF
C3 — Reference ZIP	Candidate implementation; quality checked independently	Lowest; never proof of existence

1.2 Conflicts found between sources

Table 1.2 — Source conflicts and their resolution in this baseline

Conflict	Sources	Resolution
Strain under-prediction "14 hingga 77 kali" vs "15 sampai 47 kali"	v3 §0 p. 4 vs v3 §4.4 p. 21 and AGENTS §0	Report the reference-case ratios 15×/47× with the case named (F-41)
Severity of F-06, F-12–F-16, F-19, F-20 (P1 in AUDIT-001, P0 in v3)	AUDIT-001 vs v3 §4.10	v3 governs (D-01); both recorded (Appendix C, Table C.1)
Monotonicity sign for εt vs E1/E3	ZIP `SIGNS` has −1; v3 §4.4 candidate list omits it	Neither is adopted; independent check shows violations (F-33)
Austroads equation with SF factor	v3 §4.4 shows SF; ZIP omits it	Open (HD-04, F-49)
Workflow state names	v3 Fig. 3.2 vs current spec v2.2 §24 lifecycle	Open (HD-02, F-39); current states kept
Decision IDs D-01… vs D-001…	v3 vs `docs/DECISION_LOG.md`	Prefix `V3-` proposed (Appendix G)
Python service vs "no FastAPI yet"	v3 D-04 vs existing D-001	v3 governs; D-001 marked superseded in part after approval
Commit trailers	AGENTS §6 (`Assisted-by:`) vs the repository owner's attribution rule	Open (HD-08)
"Trust Gate rejects out-of-corridor" vs "REFERRED"	ZIP `trust_gate.py` vs v3 §4.1	Specification wins; F-43

1.3 Method

1. Read-only reconnaissance: repository root, branch, commit, `git status` (including ignored files), recent log, CI workflows, test inventory, existing project instructions (none at repository root; the user-level memory notes were consulted), and the user's running development server on port 3000 (left untouched).

2. Recovery of AUDIT-001 from `docs/audit/forensic-2026-10-14/` and re-validation of each finding (Appendix C).
3. Full reading of AGENTS.md and of all 65 PDF pages via `pdftotext -layout` and `-raw` (the table of §4.10 and §1.3 needed raw order). Page images could not be rendered (no `pdftoppm`); diagrams were read from their text layer.
4. ZIP listed, hashed and extracted into an isolated temporary directory; every file read; nothing executed.
5. Independent numerical checks (`evidence/v3check.mjs`, `evidence/v3check2.mjs`; Node v24.16.0; run log `evidence/evidence-run.log`).
6. Comparative assessment, registers, this report, and an automated PDF QA pass.

1.4 Why the test suite was not re-run

AUDIT-001 ran `npm test` (226/226 pass) and `npm run lint` (clean) on this exact working tree on 2026-10-14 (its tests T-03, T-04). No file under `src/`, `server/` or `tests/` has changed since (Appendix H, T-11). Re-running would produce no new information, so the earlier results are cited. Re-running becomes mandatory with the first code change (Chapter 11).

Chapter 2 — Previous Audit Findings and Current Status

All 31 AUDIT-001 findings were re-checked against the current code. The full crosswalk with evidence and next action per finding is Appendix C.

Table 2.1 — Status of previous findings

Status	Count	Meaning for planning
CONFIRMED_OPEN	22	Fix as specified in v3 §4.10
SUPERSEDED_BY_V3	7	Requirement replaced (e.g. anti-spoofing → Trust Gate); the defective code is still shipped and must be removed
PARTIALLY_RESOLVED	1	F-15: v3 decided the source of truth and the corridors; code still Karawang-only and still calls a cloud LLM
BLOCKED	1	F-24: no Supabase credentials
RESOLVED_VERIFIED	0	—

The five original P0 findings — formula presented as PINN (F-01), fabricated telemetry (F-02), strains one to two orders too low (F-03), fake WIM training (F-04) and invented CV confidence (F-05) — are all still open. F-03 is now confirmed by two independent solvers (PyMastic as reported by v3, and the audit solver re-run here).

Chapter 3 — V3 Specification Traceability

Appendix A maps every v3 decision (D-01–D-18), role feature (W1–W7, O1–O6, E1–E7, K1–K6), module (A–H), API endpoint, prohibited claim (§1.5) and demo Definition-of-Done item (§8.4) to the current code.

Table 3.1 — Traceability summary

Group	Items	TERBUKTI	RENCANA / TARGET	DILARANG present
Decisions D-01–D-18	18	0	18	8 contradicted by shipped text (D-02, D-05, D-06, D-07, D-11, D-12, D-13, D-14)
Role features	27 (incl. admin)	2 (O3 review with reason; O5 separation of duties)	23	2 (K1 constants, K5 PDF)
Modules A–H	10 rows	0	10	—
API contracts	5 + data model	0	6	—
Prohibited claims	8	—	—	6 present, 2 not observed
Demo DoD	6	0	6	DoD-1 fails today

The parts of the current system that v3 explicitly builds on — state machine, server-side RBAC, separation of duties, audit trail, honest status vocabulary in the new domain layer, `esal.ts` with the 8.16 t axle, persisted What-If — remain sound and are the reason this baseline recommends no rewrite of the workflow layer.

Chapter 4 — Reference ZIP Technical Assessment

Appendix B lists every file with reported status, independent evidence, compatibility and disposition. Key points:

- **Safety.** 11 plain-text files, no unsafe paths, links, binaries or network/shell calls; extracted only to an isolated temporary directory. One risky default: `np.load(..., allow_pickle=True)` on a dataset file.
- **What holds up.** `mlet_strains` produces the reference strains that an independent solver reproduces (M-01–M-03); the Asphalt Institute constants, the life-index closed form, the ESAL example, the priority example and the knapsack property are correct (M-09, M-13, M-17–M-19).
- **What does not.** The surrogate feature set cannot represent the dual-tyre geometry it is trained on (F-32); two of six monotonicity constraints contradict layered-elastic results (F-33); the D-04 guardrail exists only in prose (F-35); training lacks OOD and ablation (F-34); Monte Carlo seeding and zero growth are mishandled (F-36); the number checker rejects English decimals and citation years and accepts swapped values (F-37); the Trust Gate drops out-of-corridor photos instead of referring them (F-43); YOLO is AGPL-3.0 (F-42).
- **Status.** The other team's "DIUJI" applies to their environment. In this repository every reference component is RENCANA until R1-02.

Table 4.1 — Disposition summary

Disposition	Components
INTEGRATE_REFERENCE (after R1-02)	`mlet_strains`, `life_index_years`, `priority_score`, `test_core.py` as pytest
ADAPT	`allowable_reps`, ESAL functions, knapsack, number checker, `trust_gate.py`, `gen_mlet_dataset.py`, `mc_life.py`, `slm_brief.py`, `bench_tools.py`
INVESTIGATE → team rewrite	`train_pi_surrogate.py`
DEFER	`train_yolo.py` (licence decision and local photos first)

Chapter 5 — Current-versus-Target Architecture

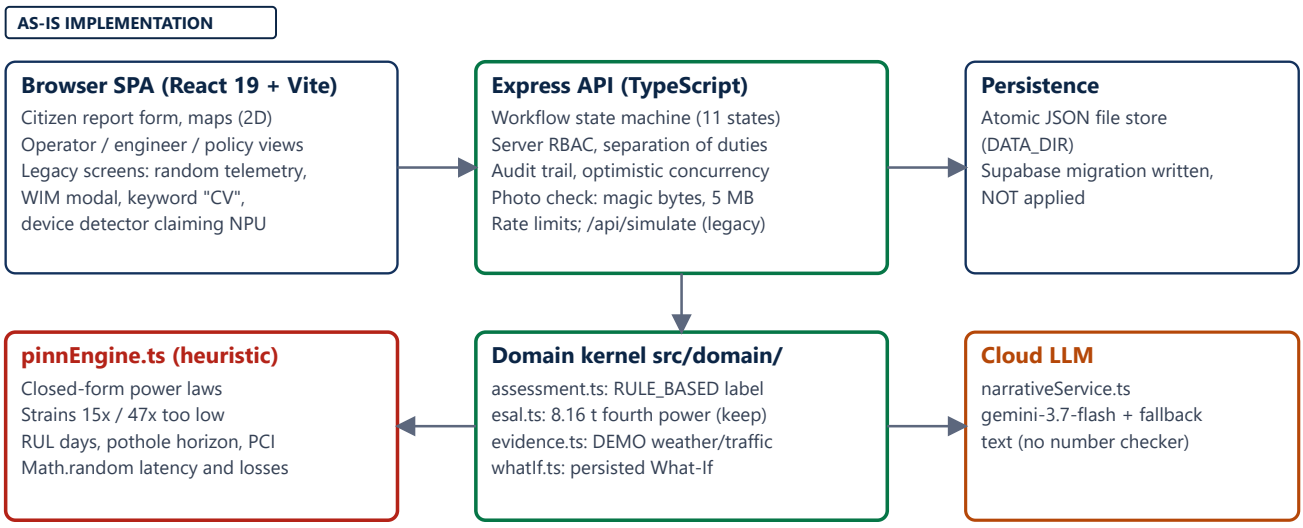


Figure 5.1 — As-is architecture (browser to Express over HTTPS/JSON; the domain kernel calls the heuristic engine). AS-IS IMPLEMENTATION, from source inspection on 2026-10-15.

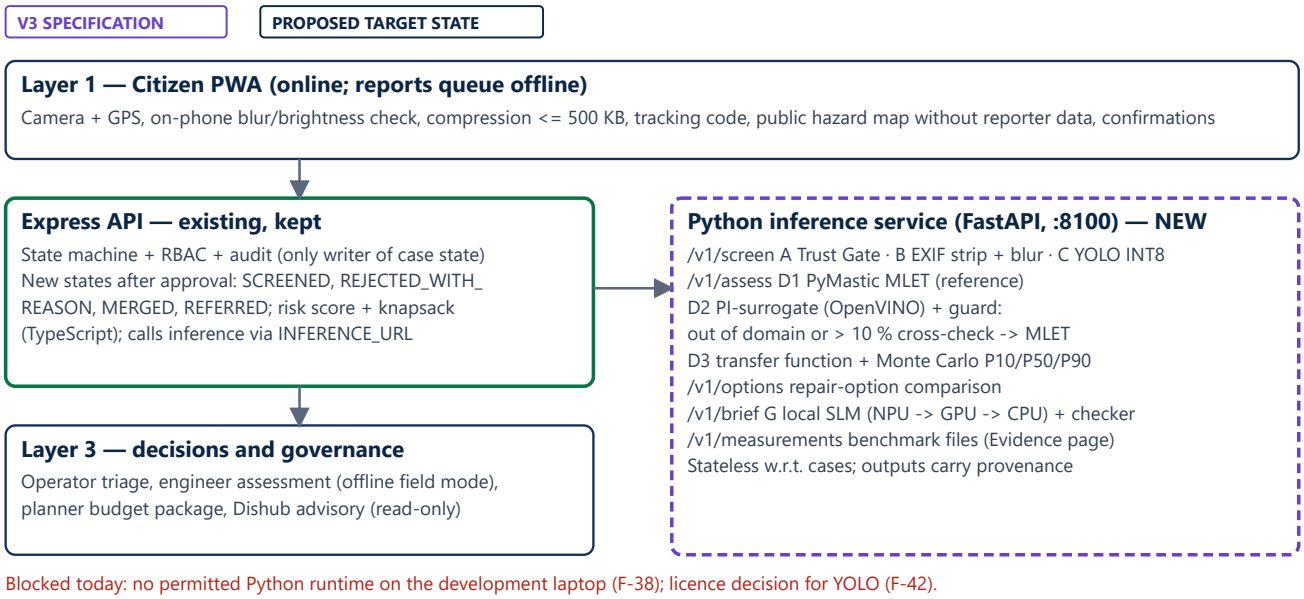


Figure 5.2 — v3 target architecture. V3 SPECIFICATION (§3.2) as PROPOSED TARGET STATE; nothing in the Python service exists yet.

5.1 Component comparison

Table 5.1 — Component-by-component comparison and disposition

Component	Current AS-IS IMPLEMENTATION	v3 target V3 SPECIFICATION	Class today	Disposition
Frontend	React 19 + Vite SPA, Leaflet/MapLibre 2D	Same + PWA/Service Worker, ID/EN	Production-capable (non-PWA)	KEEP + ADAPT

Component	Current AS-IS IMPLEMENTATION	v3 target V3 SPECIFICATION	Class today	Disposition
Workflow API	Express, state machine (11 states), RBAC, audit, optimistic concurrency, file store	Same + 4 new states, routing	Production-capable	KEEP; states INVESTIGATE (HD-02)
Identity	Server-signed sessions; credential login being added in the working tree	Real identity for internal roles (D-18)	Partial	ADAPT
Photo intake	Magic bytes, 5 MB, duplicate hint 30 m/7 d; EXIF kept	Trust Gate + EXIF strip + blur + deletion	Partial	ADAPT (keep checks, add Module A/B)
Damage classification	Keyword match, constant confidence	YOLO 4 classes OpenVINO INT8	Mock	REMOVE now; Module C later
Structural engine	`pinnEngine.ts` heuristic formula	PyMastic MLET + PI-surrogate + guard	Mock/heuristic	REPLACE (justified by F-03)
Life/deterioration	RUL days, pothole horizon, PCI projection	Indeks Umur P10/P50/P90	Heuristic, prohibited	REMOVE + build Module D3
Traffic load	Demo provider; "Virtual WIM" modal	Counts × VDF (Module E)	Demo/mock	REMOVE WIM; ADAPT provider
ESAL	`esal.ts`, 8.16 t fourth power	Same + counts × VDF	Production-capable	KEEP
Prioritisation	Workflow buckets	Transparent score + knapsack	None	INTEGRATE_REFERENCE (TypeScript)
Narrative	Cloud Gemini + fallback	Local SLM on NPU + checker	Cloud dependency	REPLACE (D-10)
Telemetry/benchmark	`Math.random`, constants, simulated benchmark	`measurements/*.json` only	Fabricated	REMOVE then build Evidence page
Device detection	Browser guesses NPU	Python `available_devices`	Fabricated	REPLACE
PDF export	Agency names, no disclaimer	Status block + disclaimer per page	Defective	ADAPT
Persistence	Atomic JSON file store; Supabase migration unapplied	Supabase or VM + backup	Partial	DEFER to HD-06
Tests	226 node tests; no reference/E2E/pytest	+ pytest + MLET reference + Playwright	Partial	ADAPT

5.2 End-to-end workflow trace

Table 5.2 — Golden path: current vs target

Step	Current path	Target path (v3 Fig. 3.2)	Gap
Citizen report	`SUBMIT` → `SUBMITTED`; photo validated (type/size) and stored with EXIF	Compressed photo, on-phone checks, tracking code	Compression, tracking code, EXIF strip
Input validation / privacy	`BEGIN_SCREENING` (SYSTEM) → rule screening, duplicate hint, OUT_OF_SCOPE for non-Karawang	`/v1/screen`: EXIF strip → Trust Gate → blur → YOLO	Entire Module A/B/C; corridor config
Damage classification	Keyword match (display only)	Versioned YOLO; confidence internal only	Model

Step	Current path	Target path (v3 Fig. 3.2)	Gap
Operator triage	`SCREENING_PASSED/NEEDS_REVIEW` → `VERIFIED` or `CLOSE_INVALID` with reason	`SCREENED` → `TRIAGED` by risk score; REJECTED_WITH_REASON, MERGED, REFERRED	Score, merge, referral, state names
Structural assessment	`assessment.ts` RULE_BASED formula, honestly labelled	`/v1/assess`: PyMastic εt/εv, P10/P50/P90, criterion	Python service
Engineering review / recommendation	`SUBMIT_INSPECTION`, `RECOMMEND_ACTION` by engineer	Same + option comparison + SLM brief	Options, SLM
Human approval	`APPROVE_ACTION` by operations manager; self-approval forbidden	Same (O5)	none — keep
Repair	`START_WORK`, `COMPLETE_WORK_ORDER` (internal, not SPK)	Same	none
Repair verification	`RESOLVE` by engineer → `CLOSE_RESOLVED`	After-photo + YOLO check + citizen confirmation	After-photo rule, confirmations

Chapter 6 — Mathematical and Scientific Verification

Appendix D holds the full register (M-01–M-25) and equations EQ-1 to EQ-7. This chapter states the conclusions.

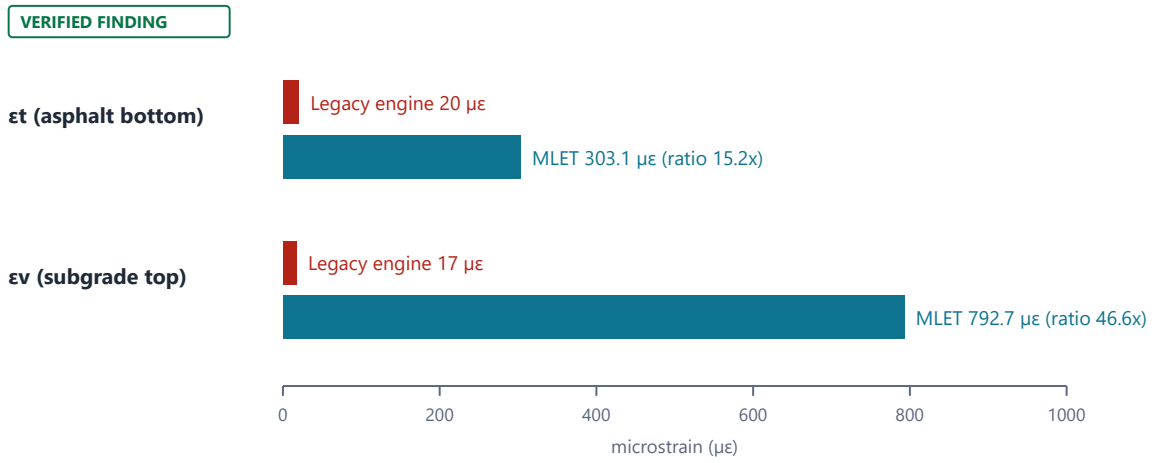


Figure 6.1 — Critical strains on the v3 reference case (10 t, 4 isolated tyres, q 0.55 MPa). Legacy engine values from AUDIT-001; MLET values from the independent audit solver, matching the PyMastic values reported in v3. VERIFIED FINDING.

6.1 Engine identity (D-02)

The current engine is a heuristic closed-form formula (M-24). The reference package contains a numerical physics solver wrapper (PyMastic) and an untrained surrogate design. The project therefore has **neither a PINN nor a trained physics-informed surrogate**. After R2 it would have a hybrid: physics solver as reference, surrogate as accelerator, guarded fallback to the solver.

6.2 MLET

The reference values are reproduced independently to 0.1 % (M-01, M-02) and elastic similarity is exact (M-03). Contact radius, sign convention and sampling depths are consistent (M-05–M-07). The dual-tyre values (227/876 με) and the PyMastic unit convention could not be reproduced without Python (M-04, M-05) and remain to be shown by R1-02.

$$a = \sqrt{\frac{10 \times 9806.65 \,/\, 4}{\pi \times 0.55}} = 119.1 \text{ mm} \qquad \text{(EQ-8)}$$

Reference-case contact radius used in M-01 (10 t axle, 4 tyres, q = 0.55 MPa).

6.3 Transfer functions and life index (D-05)

The Asphalt Institute constants match the published form (M-09). The Austroads branch is the 2004 form without the SF/RF factor shown in v3 (M-10, F-49). The criteria differ about 200–300× in N_d (M-11), so the choice must be made, cited and displayed by a human (HD-04). The life-index closed form is correct (M-13); its use in `mc_life.py` breaks for zero growth (F-36). No output may be expressed in days or as a pothole horizon.

6.4 Surrogate validation (D-04)

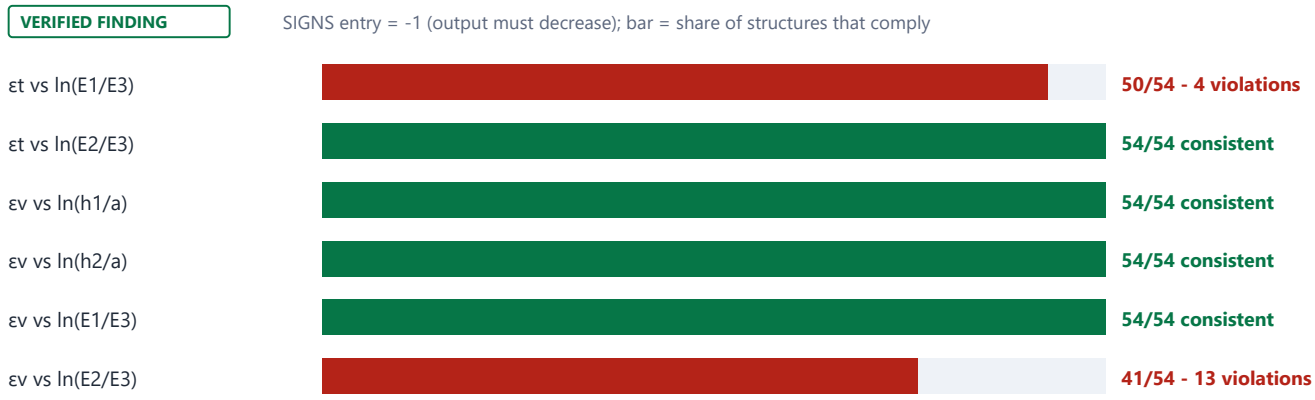


Figure 6.2 — Direction check of the six non-zero SIGNS entries: share of 54 layered-elastic structures in which a +10 % change of the feature moves the output in the required direction. VERIFIED FINDING (isolated tyre, on axis); the dual configuration remains to be checked with PyMastic.

The pipeline MLET → dataset → training → held-out validation → OpenVINO export → cross-check → fallback has no executed step. Two design defects must be fixed before data generation: the d/a ratio (F-32) and the sign table (F-33). The 10 % cross-check fallback is a target acceptance criterion with no code yet (F-35). No surrogate accuracy, FP16/INT8 deviation or timing may be cited (M-22, M-23).

6.5 Monte Carlo

Determinism is designed in (hash seed) but not executed. Required changes: common random numbers between dry and wet scenarios, seed tied to the surrogate version, zero-growth branch, provenance fields (F-36). P10/P50/P90 describe the spread of the stated input distributions, which are assumptions — they are not calibrated confidence intervals and must not be presented as such.

6.6 Other scientific claims

ESAL uses the 8.16 t standard axle in `esal.ts` (keep) while the legacy engine still uses 10 t (F-08). The VDF values in the v3 example are illustrative, not MDP 2017. The priority score reproduces its documented example; it is a policy instrument, not a learned model, and its validation target (Spearman ≥ 0.6 against blind engineer rankings) is a TARGET. Surface severity rules from photos are approximate by design (v3 §4.3). No FWD, deflection, WIM or field data exist: every structural number is at best numerically verified, never field-validated.

Chapter 7 — Data, Privacy, and Responsible AI

Table 7.1 — Responsible-AI requirements against the current build

Requirement (v3)	Current state	Gap / finding
Numerical honesty (D-11, AGENTS rule 1)	15 `Math.random` call sites in runtime UI/engine code; simulated benchmark; "Latensi <10 ms" text	F-02, F-14, F-20 — Phase 0
Provenance per output (AGENTS rule 3)	New domain layer records provenance and status (`RULE_BASED`, `DEMO`, `SYNTHETIC`); legacy screens do not	F-23; extend to inference outputs (model version, device, units, status)
Human authority (§3.5)	AI never transitions state: only `SYSTEM` screening steps, all human decisions role-guarded; separation of duties tested	Keep; inference service must not call the state machine
Privacy (D-09)	EXIF kept, no blur, no deletion job, no retention log	F-46 — R3-01, R3-03
Public-map privacy (W5)	Not verified in this baseline	Add public-response field test
Rate limit and duplicates	Rate limiting on intelligence and workflow routes; duplicate hint	Add pHash, per-device salted hash
GenAI (D-10)	Cloud Gemini call active (key on this machine is invalid, so the fallback text is served)	R4-03; "100 % offline" must never be claimed (§1.5)
Licensing (§6.5)	PyMastic Apache-2.0 (attribution needed on adoption); Ultralytics AGPL-3.0 undecided; reference ZIP has no licence	F-42, F-48 — HD-03, HD-09
GenAI disclosure (§6.6)	No disclosure sheet; trailer convention unresolved	HD-08
Prohibited claims (§1.5)	6 of 8 present	Appendix A, Table A.4

Data status: the repository holds no empirical road, deflection, traffic-count or labelled image data. Segment data are reference/demo values and must stay labelled DEFAULT or ASSUMED. PyMastic-generated data, when produced, are SYNTHETIC and must never be presented as field data.

Chapter 8 — Findings Register and Risk Prioritisation

Appendix E lists 51 findings: the 31 carried over (v3 priorities) and 20 new ones (F-32–F-51).

Table 8.1 — Open findings by severity

Severity	Carried over	New	Total
P0	18	1	19
P1	9	6	15
P2	4	10	14
P3	0	3	3

Table 8.2 — Top risks for the next milestone

Rank	Risk	Findings	Mitigation
1	False or unmeasured claims seen by judges	F-01, F-02, F-05, F-06, F-14, F-16, F-19, F-20	Phase 0, CI grep test
2	Physics path cannot start: no Python	F-38	HD-01
3	Personal data in stored photos	F-46	R3-01 now, Module B later
4	Surrogate trained on a flawed design	F-32, F-33, F-35	R2-01, R2-02, R2-04 before training
5	Licence exposure from YOLO	F-42	HD-03 before integration
6	State-machine change breaks the verified backbone	F-39	HD-02, mapping first, existing tests untouched

Chapter 9 — Integration and Migration Strategy

9.1 Boundaries

1. **Express stays the system of record.** It owns cases, RBAC, audit and every state transition. The Python service is stateless with respect to cases; Express calls it via `INFERENCE_URL` (default `http://localhost:8100`) and stores the response as provenance-tagged evidence.
2. **Python owns physics and models.** MLET, transfer functions, Monte Carlo, surrogate and guard, Trust Gate, privacy, YOLO, SLM, benchmarks live in `inference/` with pytest.
3. **TypeScript owns policy logic.** Priority score, budget optimiser and ESAL × VDF go to `src/domain/intelligence/` next to `esal.ts`, so the operator and planner screens stay deterministic and testable in the existing suite.
4. **Fail closed.** If the inference service is down, the UI shows "layanan inferensi tidak tersedia" and no number — never a fallback formula.

9.2 Adoption rules for reference files

Table 9.1 — Adoption rules

Rule	Detail
Reproduce first	A reference file is adopted only after its tests run on the team's machine (R1-02)
Provenance header	Adopted files carry origin (v3 package, sha256), licence and "adapted by"
Team-written science	Training, transfer functions, Monte Carlo, evaluation and benchmarks are written or rewritten by team members (AGENTS §6); AI may review and write tests
No silent behaviour change	Legacy tests that encode physically wrong behaviour are marked legacy with approval
Dependencies	`requirements.txt` with pins; Ultralytics only after HD-03
Data and models	Gitignored; only hashes, cards and `models/registry.json` committed

9.3 Proposed change sequence

Table 9.2 — Change sequence (each line one or more small PRs)

Order	Change	Files (indicative)	Gate
1	`no-fake-claims.test.ts` (failing first)	`tests/unit/`	G-0
2	Remove random telemetry/attribution; "belum diukur"	`pinnEngine.ts`, `CommunityReportsDashboard.tsx`, `LeafletMap.tsx`, `Navbar.tsx`, `FieldEngineerTools.tsx`, dashboards	G-0
3	Delete WIM modal/scraper and IR export routes	`features/wim/*`, `services/wim/*`, `openvino*` server files, routes	G-0
4	CV "belum diklasifikasi"; device "tidak diketahui"; verdict text removal	`cvDamageClasses.ts`, `deviceDetector.ts`, `pinnEngine.ts`	G-0
5	PDF disclaimer; README/package; status labels	`pdfGenerator.ts`, `README.md`, pages	G-0
6	Corridor config (Narogong + Karawang)	`config/corridors/`, `screening.ts`	G-0, HD-07

Order	Change	Files (indicative)	Gate
7	DECISION_LOG merge	`docs/DECISION_LOG.md`	approval
8	`inference/` + `/v1/assess` + Express adapter	new folder, `server/`	G-1
9	Surrogate pipeline and guard	`inference/training/`, `inference/core/`	G-2
10	Screening, privacy, detector; new states	`inference/`, `src/domain/workflow/`	G-3, G-4
11	SLM, benchmarks, Evidence page	`inference/genai/`, `inference/bench/`, UI	—
12	Role features, Playwright, deployment	various	G-5

Chapter 10 — Development Roadmap

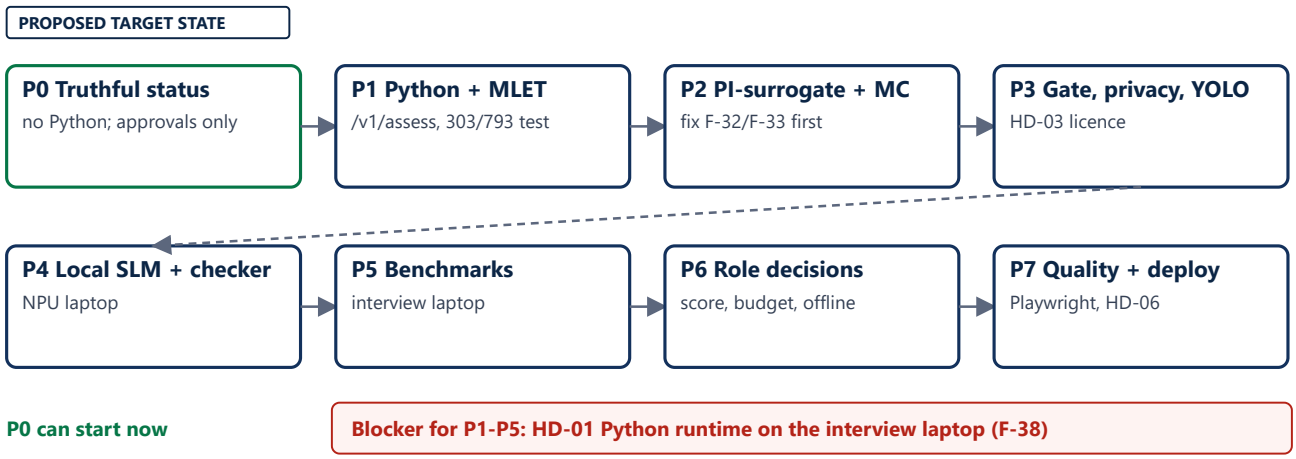


Figure 10.1 — Roadmap phases, their dependencies and blockers. PROPOSED TARGET STATE.

Appendix F holds the 52 tasks (R0-01 … R7-04) with source requirement, gap, change, dependencies, tests, acceptance criteria, priority, track and size, plus the approval gates G-0…G-5. Recommended first phase: **M-P0 "Truthful status"** (R0-01–R0-10) — it needs no Python, removes five of the six prohibited claims still on screen (the day-based life and horizon fields go with R1-06) and makes V3-DOD-1 pass. Phases 1–5 are blocked until HD-01 provides a Python runtime on the interview laptop. The v3 seven-day emergency plan cannot be met as written while that blocker is open; calendar estimates are therefore withheld.

Chapter 11 — Test Plan and Definition of Done

Table 11.1 — Test plan per phase

Phase	New tests	Gate commands	Definition of done
0	`no-fake-claims.test.ts` (grep `Math.random` in runtime paths; forbidden phrases; payload equality); route 404 tests; PDF snapshot; corridor screening test	`npm test`, `npm run lint`	M-P0 exit criteria (Appendix F.5)
1	pytest: 303/793 ±1 %, 274/663, ×3 similarity, invalid input raises; contract tests `/v1/assess`; Node adapter test; cross-check vs audit solver on axis	+ `pytest`	Engineer UI shows PyMastic strains, criterion, "indikatif, bukan desain"; no legacy life fields
2	Dataset card check; sign-check report; metrics schema (random, OOD, similarity, violation %, ablation); guard tests; MC: P10 ≤ P50 ≤ P90, determinism, dry ≥ wet per sample, g = 0 finite	+ `pytest`	`models/pi_surrogate_metrics.json` complete; fallback flagged
3	EXIF-strip test; gate branch tests; blur recall report; deletion-log test; "no silent drop"; public-response privacy test	+ `pytest`	FAR/FRR, recall and mAP per class/condition stored
4	50-case checker suite (ID and EN, citations, swapped values); network-off test	+ `pytest`	Pass rate and tokens/s per device stored
5	`measurements/*.json` schema test	+ benchmark script on the interview laptop	Evidence page reads only `/v1/measurements`
6	Score example 83.3; weight-change audit; DP ≥ greedy; budget curve monotone; option monotonicity; merge/SLA/REFERRED tests	all	§3.4 acceptance criteria
7	Playwright golden path	all + Playwright	v3 §8.4 DoD all ticked

Non-negotiable for every phase: no existing test deleted or weakened; tolerances of reference tests never widened to make them pass; failures recorded with their output.

Chapter 12 — Readiness and Unresolved Decisions

Table 12.1 — Readiness

Item	Ready?	Note
Phase 0 implementation	Ready after approval	Pure TypeScript; change plan per PR (> 3 files)
Phase 1–5 implementation	Blocked	F-38 / HD-01
Reference code adoption	Not ready	Reproduce first (R1-02); fix F-32–F-37, F-43
Surrogate training	Not ready	HD-05, R2-02 sign check
YOLO integration	Not ready	HD-03 licence; local photos
State-machine extension	Needs approval	HD-02
Persistence/deployment	Needs decision	HD-06
Pitch claims	Restricted	Only TERBUKTI items; see Chapter 0 list of things that must not be claimed

Open human decisions HD-01 to HD-09 are listed with options in Appendix G, Table G.2. This baseline did not take any of them.

Readiness statement. The project is not v3-compliant. It has a sound, tested workflow core and a clear, evidence-based path. The single most valuable next action is Phase 0; the single biggest risk is the missing Python runtime.

Appendix A — V3 Requirement Traceability Matrix

Document PAVEMENT-PINN-V3-BASELINE-001 · Version 1.0 · 2026-10-15. Page numbers are the PDF's own footers ("Halaman N / 65") of *PAVEMENT-PINN v3 Dokumen Induk* (sha256 `1cdc4229…b6ab`).

Claim status of the current repository (v3 §1.1): TERBUKTI runs in this repository with evidence; TARGET a measurable threshold not yet measured; RENCANA not built; DILARANG the current build shows a claim v3 forbids. Requirement IDs are v3's own where it has them (D-xx, W1…K6); other rows get stable IDs: `V3-MOD-*` module criteria, `V3-API-*` endpoints, `V3-PC-*` prohibited claims (§1.5), `V3-DOD-*` demo Definition of Done (§8.4), `V3-RAI-*` responsible-AI items, `V3-BEN-*` benchmark protocol.

A.1 Decisions D-01 to D-18 (§1.3, pp. 7–8)

Table A.1 — Decision traceability

ID	Decision (abridged)	Existing implementation (evidence)	Status	Evidence needed	Acceptance criterion
D-01	v3 document governs over blueprint v1, finalist doc v2 and spec v2.2; copy decisions to `docs/DECISION_LOG.md`	`docs/DECISION_LOG.md` still cites spec v2.2 as top source (l.4-6); IDs collide (F-40)	RENCANA	Merged log approved by team	DECISION_LOG lists V3-D-01…18 and marks superseded entries
D-02	"PINN" = physics-informed neural surrogate; say it is not a PDE-residual PINN	Legacy UI/README still say PINN solver (F-01); new domain layer honest (`assessment.ts:4-8,17`)	DILARANG on legacy screens	grep CI test	No "Physics-Informed Neural Network/PINN Solver" text outside roadmap-labelled copy
D-03	Corridor 1 Narogong, corridor 2 Karawang, corridors as configuration	Karawang hard-coded (`karawangPilot.ts`); Narogong reports flagged OUT_OF_SCOPE (`screening.ts:83`)	RENCANA	`config/corridors/*.json`, test	Narogong report not OUT_OF_SCOPE; adding a corridor needs no code change
D-04	MLET via PyMastic as reference; PI-surrogate for batches; fallback when out of domain or > 10 % cross-check	None (heuristic engine)	RENCANA	`/v1/assess` tests, guard tests (F-35)	303/793 ±1 %; guard tests pass
D-05	Selectable, cited transfer function; Indeks Umur P10/P50/P90 in years, "indikatif, bukan desain"; no pothole horizon	RUL days and pothole horizon still computed (F-07)	DILARANG (horizon) / RENCANA (life index)	Criterion decision HD-04	UI shows criterion + label; no day-based field in any API
D-06	No PCI projection; record observed condition; ASTM D6433 only from a distress survey	PCI projection in `pinnEngine.ts` (F-07)	DILARANG	grep/API test	No projected PCI field
D-07	YOLO 4 classes (RDD2022 + local), OpenVINO INT8, confidence only from a versioned model	Keyword classifier with constant confidence (F-05)	DILARANG	Model registry + test	Confidence displayed only when `modelStatus=REAL` and version set
D-08	Photo Trust Gate with measured FAR/FRR	Magic bytes, 5 MB, duplicate hint 30 m/7 d (`photoValidation.ts`, `screening.ts`)	RENCANA	ROC + FAR/FRR report	FAR/FRR stored per check
D-09	Blur faces/plates, strip EXIF server-side before storage, raw photo deleted ≤ 7 days	Not done; EXIF kept (F-46)	RENCANA	Tests + deletion log	See R3-01
D-10	Local SLM on NPU, structured input, numbers by code, checker, template; cloud calls off	Cloud Gemini call (`narrativeService.ts:9`) with fallback	RENCANA	`/v1/brief` tests; network-off demo	No outbound LLM call; checker pass rate measured
D-11	All performance numbers from `measurements/*.json`; otherwise "belum diukur"	Random/constant telemetry (F-02, F-14)	DILARANG	`no-fake-claims.test.ts`	Evidence page reads only `/v1/measurements`
D-12	Remove "Virtual WIM"; traffic load from classified counts × MDP 2017 VDF (normal vs actual)	Demo traffic provider (F-25); WIM modal (F-04)	DILARANG (WIM) / RENCANA	Counts + VDF table with source	Recomputable from the count form

ID	Decision (abridged)	Existing implementation (evidence)	Status	Evidence needed	Acceptance criterion
D-13	Dishub panel advisory only; no rerouting orders or dynamic tonnage limits	Advisory panel absent; legacy engine text still issues a dynamic tonnage limit and a "Tindakan langsung" order (`pinnEngine.ts:279`)	DILARANG (legacy text) / RENCANA (panel, P2)	UI review + grep	No action buttons; no tonnage limit or order text
D-14	Material recommendation → repair-option comparison (patch, AC-WC, PMA) with editable moduli; remove "PMA ≥ 8.5 cm"	PCI look-up overlay (F-18)	DILARANG	`/v1/options` test	Thicker/stiffer option never gives lower life
D-15	Quality audit → neutral early-damage investigation flag (≥ 3 reports + engineer confirmation)	Not present	RENCANA (P2)	Word-list test	Flag text free of accusatory words
D-16	2D map + strain section from real MLET grid; remove 3D twin and pore-pressure animation	2D section from a div with an unused base layer (F-29)	RENCANA	Visual review	Section geometry equals model inputs
D-17	Segment stores managing authority; automatic routing; citizen sees responsible agency	Not present	RENCANA	Segment config + routing test	Out-of-authority report becomes REFERRED with target agency
D-18	Citizen PWA live on a public URL; Supabase applied or one VM with daily backup; real identity for internal roles	GitHub Pages workflow exists for the static demo; file store; credential login being added in the uncommitted working tree (D-034)	RENCANA	HD-06 decision, deployment log	Public URL serving the pilot; backups verified

A.2 Role features (§3.4, pp. 15–17)

Table A.2 — Role feature traceability

ID	Prio	Feature	Existing implementation (evidence)	Status	Acceptance criterion (v3)
W1	P0	3-step report from browser camera, GPS accuracy, compression to ≤ 500 KB	Report flow exists (`ReportRoadIssueCard.tsx`, server `POST` with photo validation); no client compression to 1280 px	RENCANA (compression)	Median time ≤ 60 s on 10 testers; median upload ≤ 500 KB
W2	P1	On-phone blur/brightness check	None	RENCANA	≥ 90 % of 30 synthetic blurred photos rejected on the phone
W3	P1	Offline queue (Service Worker + IndexedDB, idempotency key)	No service worker	RENCANA	Airplane mode → online: sent exactly once
W4	P0	Tracking code + public status page	No tracking code (grep: none)	RENCANA	Status equals state machine (integration test)
W5	P0	Public hazard map with verified points only, no reporter data	Map exists; public-response privacy not verified in this baseline	RENCANA (needs check)	No personal field in public API responses (test)
W6	P1	Citizen confirmation "masih ada"/"sudah diperbaiki"	None	RENCANA	Confirmation raises component C (unit test)
W7	P1	ID/EN, large buttons, WCAG AA	Indonesian UI; no i18n framework	RENCANA	Lighthouse accessibility ≥ 90

ID	Prio	Feature	Existing implementation (evidence)	Status	Acceptance criterion (v3)
O1	P0	Queue ordered by risk score with contribution breakdown; editable weights logged	Queue by workflow bucket; no score	RENCANA	Order changes with weights; change audited
O2	P1	Merge duplicates (≤ 30 m, ≤ 7 days + pHash)	Duplicate hint 30 m/7 d exists (`screening.ts`); no merge, no pHash	RENCANA (partial)	≥ 18 of 20 synthetic pairs merged
O3	P0	NEEDS_REVIEW with reasons; accept/reject with reason; nothing dropped silently	`NEEDS_REVIEW`, `REVIEW_ACCEPT`, `CLOSE_INVALID` (reason) exist (`stateMachine.ts:49-51`)	TERBUKTI (state machine, tests) — new REJECTED_WITH_REASON not yet	"No silent drop" test
O4	P1	Assign + SLA counter per hazard level	Assignment exists (`REQUEST_INSPECTION`, deadline validation `stateMachine.ts:213`); no SLA per hazard	RENCANA (SLA)	Late SLA marked red
O5	P0	Approve work order; recommender ≠ approver; "bukan SPK"	Separation of duties and internal work order verified by audit (S-01, D-012/D-013)	TERBUKTI	"self-approval forbidden" test stays green
O6	P1	Authority routing → REFERRED	None	RENCANA	Out-of-authority report → REFERRED
E1	P0	Offline field mode on the laptop (local YOLO, PyMastic, surrogate, SLM; sync with version check)	Version check (optimistic concurrency) exists; no local inference	RENCANA	Wi-Fi off: full inspection works
E2	P0	Correct detections → labels with source model version (Module H)	None	RENCANA	Correction appears in export
E3	P0	Structural assessment: εt/εv from PyMastic, P10/P50/P90, sensitivity, section, provenance	`assessment.ts` RULE_BASED heuristic with honest label; What-If persisted	RENCANA	εt/εv equal PyMastic ±1 %; P10 ≤ P50 ≤ P90; "indikatif" visible
E4	P1	Compare repair options (life + cost)	What-If exists for engine inputs only	RENCANA	Monotonicity test
E5	P0	Recommendation + SLM summary, engineer edits and approves	`RECOMMEND_ACTION` exists; summary from cloud/fallback	RENCANA (SLM)	All numbers pass checker
E6	P1	Repair verification with after-photo	`RESOLVE` by engineer exists; no after-photo requirement found	RENCANA	Case cannot be verified without after-photo
E7	P2	Early-damage flag	None	RENCANA	No accusatory words
K1	P0	Corridor KPI dashboard from real data only	Policy view with ILUSTRATIF notice; some constants (F-22)	DILARANG where constants shown	Every number traceable to a query (test)
K2	P1	Budget package optimisation (knapsack DP + greedy)	None (reference code available, M-19)	RENCANA	DP ≥ greedy on 100 random cases
K3	P1	Budget scenario curve	None	RENCANA	Curve non-increasing in budget
K4	P1	Coverage-equity map	Report density module exists (`reportDensity.ts`)	RENCANA (partial)	Segments without reports still shown

ID	Prio	Feature	Existing implementation (evidence)	Status	Acceptance criterion (v3)
K5	P0	Policy summary + PDF with status block and disclaimer on every page	PDF with agency names, no disclaimer (F-19)	DILARANG	PDF snapshot test
K6	P2	Dishub load-enforcement advisory panel	None	RENCANA	No action buttons
ADM	P1	Corridor/segment config, model registry, retention settings, audit export	Audit trail exists; no registry/retention UI	RENCANA	—

A.3 Modules, contracts, governance

Table A.3 — Module, API and governance traceability

ID	Section/page	Requirement	Existing implementation	Status	Acceptance criterion
V3-MOD-A	§4.1 pp. 19–20	Trust Gate checks 1–8 with ROC-calibrated thresholds; FAR ≤ 5 %, FRR ≤ 5 % (blur/dark), screen-photo FAR ≤ 10 %	Magic bytes + size + duplicate hint	TARGET	Measured FAR/FRR reported as-is
V3-MOD-B	§4.2 p. 20	Face/plate blur (OpenVINO), EXIF strip → blur → store → delete raw ≤ 7 days; blur recall ≥ 95 % on 50 photos	None (F-46)	TARGET	Recall measured; deletion log
V3-MOD-C	§4.3 pp. 20–21	YOLO nano/small, split by location/day, ≥ 100 local test photos, mAP per class/condition, FP32 vs INT8, kappa severity	None	TARGET	Metrics file per class and condition
V3-MOD-D1	§4.4 p. 21	MLET via PyMastic, q in kPa, a-normalised, iteration 20, dual tyres at x = 0, d/2, d	None in repo; reference independently cross-checked (M-01–M-03)	RENCANA	303/793 ±1 %; 274/663; similarity
V3-MOD-D2	§4.4 pp. 22–23	PI-surrogate (dimensionless I/O, sign-checked monotonicity loss, guardrail); validation: random 20 %, OOD, similarity, violation %, ablation	None; reference has defects F-32–F-35	TARGET	Metrics JSON with all five validations
V3-MOD-D3	§4.4 pp. 23–24	Transfer function + life index + Monte Carlo 2 000 samples, deterministic seed, Spearman tornado	None	RENCANA	P10 ≤ P50 ≤ P90; deterministic; physical direction
V3-MOD-E	§4.5 pp. 24–25	ESAL = Σ counts × VDF (normal/actual), MDP 2017 CESA, P10–P90	`esal.ts` fourth-power law (8.16 t) only	RENCANA	Recomputable from forms; constants with sources
V3-MOD-F	§4.6 pp. 25–26	Transparent score (weights 0.5/0.3/0.1/0.1); blind ranking Spearman ≥ 0.6; knapsack	None	TARGET	Example 83.3 reproduces; ρ measured

ID	Section/page	Requirement	Existing implementation	Status	Acceptance criterion
V3-MOD-G	§4.7 p. 26	SLM INT4 on NPU → GPU → CPU; placeholders; checker; template after 2 fails; label "Disusun AI..."; no cloud	Cloud Gemini	RENCANA	Pass rate on 50 cases; tokens/s per device
V3-MOD-H	§4.8 p. 26	Active learning; `models/registry.json`; promote only if fixed local test mAP does not drop	None	RENCANA	Registry committed; promotion rule tested
V3-API-1	§4.9 p. 27	`POST /v1/screen` (multipart) → trust, privacy, detections, model versions, device	None	RENCANA	Contract test
V3-API-2	§4.9 p. 27	`POST /v1/assess` → mlet, life_years, damage_used, sensitivity, surrogate {version, in_domain, crosscheck_err}, status_label, device	Express `POST` simulations/assessments (RULE_BASED)	RENCANA	Contract test + Express adapter
V3-API-3	§4.9 p. 27	`POST /v1/options`	None	RENCANA	Contract test
V3-API-4	§4.9 p. 27	`POST /v1/brief`	`POST` narrative (Gemini)	RENCANA	Contract test
V3-API-5	§4.9 p. 27	`GET /v1/measurements` — sole source of the evidence page	None	RENCANA	UI reads only this
V3-DATA	§4.9 p. 27	Tables corridors, segments (authority, layers with source), reports (tracking code, salted device hash), photos, trust_checks, detections, assessments, confirmations, priority_snapshots, traffic_counts, measurements	Migration has 17 tables (not applied); different normalisation (`report_cases`, `pinn_predictions`)	RENCANA	Schema decision recorded (HD-06)
V3-BEN	§5.4 pp. 30–31	Benchmark protocol: fixed model hash, OpenVINO version, `available_devices`, driver, 50 warm-up, ≥ 1 000 runs, cold start, p50/p95/p99, throughput, memory, deviation vs FP32, power or "tidak diukur"	Simulated benchmark (F-14)	DILARANG (simulated values)	`measurements/*.json` with all fields
V3-RAI-1	§6.1 p. 32	Risk register mitigations (reporter safety warning before camera, equity, no gamification, ...)	Not present	RENCANA	Warning text shown before camera
V3-RAI-2	§6.2 pp. 32–33	Data lifecycle: rejected reports kept 30 days then deleted; deletion via tracking code ≤ 7 days	None	RENCANA	Retention job tests
V3-RAI-3	§6.3 p. 33	Model cards (YOLO, scene classifier, PI-surrogate, SLM)	None	RENCANA	Card per registry entry
V3-RAI-4	§6.5 p. 33	Licences: PyMastic Apache-2.0 attribution; Ultralytics AGPL-3.0 decision; RDD2022; SLM licence	Not applicable yet	RENCANA	HD-03 decided before integration
V3-RAI-5	§6.6 p. 34	Honest GenAI disclosure; commits marked `Assisted-by:` (AGENTS §6)	Not present	RENCANA	Disclosure sheet; trailer convention agreed (see HD-08)

A.4 Prohibited claims (§1.5, p. 9) and demo Definition of Done (§8.4, p. 39)

Table A.4 — Prohibited claims: where the current build still violates them

ID	Prohibited claim	Present in current build? (evidence)	Status
V3-PC-1	"PINN that solves Burmister" / "FEM-equivalent in < 10 ms"	README l.1; legacy screens; "Latensi <10 ms" in `demoAccounts.ts:128`	DILARANG present
V3-PC-2	Latency, power, NPU use or quantisation accuracy not from a measurement file	`pinnEngine.ts:217`, `simulatedBenchmark.ts`, `deviceDetector.ts:15`	DILARANG present
V3-PC-3	"Validated with FWD/WIM/field data"	WIM-station correlations in `LeafletMap.tsx:362`; FWD badge (F-16)	DILARANG present
V3-PC-4	"Jalan ini aman" / "melebihi batas ijin" from model output	`pinnEngine.ts:129` ("melebihi batas ijin"), `:132` ("batas elastis aman"), `:170` ("OPERASIONAL AMAN")	DILARANG present
V3-PC-5	Remaining life in days, pothole horizon, APBN savings	`pinnEngine.ts:90-112,188-201`	DILARANG present
V3-PC-6	Data from/endorsed by Bina Marga, PUPR, Kemenhub, BMKG or Intel	`pdfGenerator.ts:34,65,163`	DILARANG present
V3-PC-7	"Detects rogue contractors", "corruption"	Not found in this baseline's sweep	Not observed
V3-PC-8	"100 % offline" for the whole system	Not found in this baseline's sweep	Not observed

Table A.5 — Demo Definition of Done (§8.4) against the current build

ID	Criterion	Current state	Blocking tasks
V3-DOD-1	No number on screen from a constant or `Math.random` (grep test)	Fails: 15 `Math.random` call sites in 4 runtime files	R0-02
V3-DOD-2	Phone photo → gate → blur → versioned detection → queue with explainable score	Not possible: no gate, blur, model or score	R3-*, R6-01
V3-DOD-3	εt/εv from PyMastic, life index P10/P50/P90, criterion, "indikatif"	Not possible: no Python service	F-38, R1-, *R2-*
V3-DOD-4	Wi-Fi off: assess and write summary on NPU	Not possible	R4-*, R6-03
V3-DOD-5	Evidence page with real benchmarks (device, OpenVINO version, date)	Not possible	R5-*
V3-DOD-6	Local backup recording of the demo	Team task	R7-04

Appendix B — Reference Code Assessment (pavement_pinn_v3_code.zip)

Document PAVEMENT-PINN-V3-BASELINE-001 · Version 1.0 · 2026-10-15.

B.1 Handling of the archive

- **Isolation.** The ZIP (17 199 bytes, sha256 `664c6a7f…2b88`) was listed with `unzip -lv` before extraction: 11 entries, all under `pavement_pinn_v3_code/`, no absolute paths, no `..` components, no symbolic links, no binaries, no model or dataset files. It was extracted into the session's temporary scratch directory, **never into the repository**. Hashes of the archive, the v3 PDF, `AGENTS.md` and of every extracted file are in `evidence/reference-inputs.txt`.
- **No execution.** No reference file was executed. Python is blocked on this machine (F-38). All statements below come from reading the code and from the independent checks in Appendix D.
- **Content vs. PDF.** The ZIP files are the code printed in v3 Appendix B (B.1–B.8). B.9 (sign check) exists only in the PDF as a snippet; the `verify_signs` helper referred to in `train_pi_surrogate.py:32` is not in the ZIP.
- **Dependencies declared** (README only, no pins): Python 3.11+, numpy, scipy, opencv-python, PyMastic (git clone, Apache-2.0); for AI modules torch, openvino, nncf, openvino-genai, optimum[openvino], ultralytics.
- **Unsafe operations found:** none. No network calls, no shell execution, no deletion. File writes are limited to `--out` paths, `models/`, `runs/` and `measurements/`. `np.load(..., allow_pickle=True)` in `train_pi_surrogate.py:65` should be `allow_pickle=False` once the `ranges` string field is stored differently (loading an untrusted `.npz` with pickle enabled can execute code).

B.2 Inventory and disposition

Table B.1 — Reference implementation inventory

Reference component	Source file (lines)	Intended module	Reported status (v3/README)	Independently reproduced here?	Compatibility with this repository	Disposition
`mlet_strains` (PyMastic wrapper, dual-tyre superposition)	`pavement_core.py` 8-39	D1	DIUJI	Reference strains 303/793 and 274/663 με and ×3 similarity reproduced with the independent audit solver (M-01–M-03); dual case not reproduced (M-04)	New Python service; nothing equivalent exists (current engine is a heuristic)	INTEGRATE_REFERENCE after R1-02 passes; replace `assert` (F-51)
`allowable_reps` (AI, AUS)	`pavement_core.py` 42-58	D3	DIUJI	AI constants match Huang; AUS omits SF/RF, edition unverified (M-09, M-10)	No transfer function exists in the repo (only text thresholds, F-06)	ADAPT: criterion choice and citation first (HD-04, F-49)
`life_index_years`	`pavement_core.py` 60-68	D3	DIUJI	Closed form verified (M-13)	Replaces RUL/horizon (F-07)	INTEGRATE_REFERENCE
`esal_per_day`, `design_esal`	`pavement_core.py` 71-77	E	DIUJI	Example 1 360 / 2 930 reproduced (M-17)	Repo `esal.ts` already has the 8.16 t fourth-power law (keep); this adds counts × VDF	ADAPT: port to TypeScript next to `esal.ts` or call from Python; VDF tables from MDP 2017 required
`priority_score`	`pavement_core.py` 80-94	F	DIUJI	Example 83.3 reproduced (M-18)	No risk score exists; queue is bucketed by state	INTEGRATE_REFERENCE — better in TypeScript (`src/domain/`) because the operator queue and audit log live there; weights editable + audited
`knapsack`, `greedy`	`pavement_core.py` 96-120	F (K2)	DIUJI	Property DP ≥ greedy reproduced on 1 000 cases (M-19)	Nothing exists	ADAPT: unit scaling (F-45); TypeScript or Python both fine
`check_brief`, `numbers_in`	`pavement_core.py` 123-139	G	DIUJI	Good/bad examples reproduced; edge-case defects (M-20, F-37)	Repo uses cloud Gemini with fallback text	ADAPT: positional slot check, locale, citation years
`test_core.py`	1-43	D, E, F, G	"ALL TESTS PASSED"	Not run (BLOCKED, F-38); its assertions were re-derived independently where possible	Becomes `inference/tests/test_core.py` (pytest)	INTEGRATE_REFERENCE as pytest; keep the 303/793 tolerance (do not widen)
`trust_gate.py` (blur, brightness, pHash, haversine, decide)	1-44	A	DIUJI	Not run; logic read; thresholds uncalibrated; REFERRED path missing (F-43)	Repo has magic-byte, size and 30 m / 7 day duplicate hint (`photoValidation.ts`, `screening.ts`) — keep those	ADAPT

Reference component	Source file (lines)	Intended module	Reported status (v3/README)	Independently reproduced here?	Compatibility with this repository	Disposition
`gen_mlet_dataset.py`	1-48	D2 data	DIUJI (small n)	Not run; LHS ranges match v3 §4.4; silent NaN drop (F-44); d/a not stored as a feature (F-32)	Offline tool, no repo coupling	**ADAPT**
`mc_life.py`	1-44	D2/D3	DIUJI	Not run; g = 0 defect, seed design (F-36)	Needs a trained predictor	**ADAPT**
`train_pi_surrogate.py`	1-101	D2	BELUM DIUJI	Not run; feature defect (F-32), SIGNS defect (F-33), validation gaps (F-34), no guardrail (F-35), `allow_pickle=True`	Offline tool	**INVESTIGATE** → rewrite by the team (AGENTS §6: scientific core written by team members)
`train_yolo.py`	1-42	C	BELUM DIUJI	Not run; split-by-location not implemented (F-50); AGPL licence (F-42)	Offline tool; model consumed by `/v1/screen`	**DEFER** until HD-03 (licence) and photos exist
`slm_brief.py`	1-51	G	BELUM DIUJI (helpers tested)	Not run; checker edge cases (F-37); NPU fallback only in a comment	Replaces `narrativeService.ts` (Gemini)	**ADAPT** after R4 hardware check
`bench_tools.py`	1-37	Evidence page	BELUM DIUJI	Not run; NPU dynamic-shape risk (F-47); protocol fields match v3 §5.4 except memory and throughput	Feeds `/v1/measurements`	**ADAPT**
`README.md`	1-9	—	—	Read	—	Fold into `inference/README.md`

B.3 What "DIUJI" means and why it is not inherited

v3 Appendix B states that "DIUJI" files were run with tests in a Python 3.13 + PyMastic + NumPy/SciPy/OpenCV environment while the document was prepared. That run is the other team's evidence, not this repository's. Under v3 §1.1 a claim becomes TERBUKTI only when it runs **in the team's repository** with stored evidence. Until R1-02 produces a log on the team's own Intel machine, every row above stays at most "independently cross-checked" (for the rows with a REPRODUCED check in Appendix D) and the rest stay "not verified".

B.4 Integration boundary (proposed)

Table B.2 — Where each adopted piece should live

Piece	Location	Language	Called by	Contract
MLET, transfer functions, life index, Monte Carlo, surrogate guard	`inference/core/`	Python	`/v1/assess`, `/v1/options`	v3 §4.9 JSON, every output with provenance + status label
Trust Gate, EXIF strip, blur, YOLO	`inference/core/`, `inference/privacy/`	Python	`/v1/screen`	v3 §4.9
SLM brief + checker	`inference/genai/`	Python	`/v1/brief`	v3 §4.9
Benchmarks	`inference/bench/`	Python	CLI → `measurements/*.json`; `/v1/measurements`	v3 §5.4
Priority score, budget optimiser, ESAL × VDF	`src/domain/intelligence/` (beside `esal.ts`)	TypeScript	Express routes, operator/planner views	Deterministic, isomorphic, audited weight changes
Workflow transitions	existing `src/domain/workflow/stateMachine.ts`	TypeScript	unchanged	AI never calls `applyCommand`; Express stores inference results as evidence only

The Express API stays the only writer of case state (AGENTS rule 4). The Python service is stateless with respect to cases: it computes and returns; Express persists and audits.

Appendix C — Previous Audit Crosswalk (F-01 to F-31)

Document PAVEMENT-PINN-V3-BASELINE-001 · Version 1.0 · 2026-10-15 · Repository commit `4a9b4fc` (branch `feature/m1-report-ownership`, working tree with another session's uncommitted demo-authentication changes, unchanged since the previous audit).

Source of the original findings: PAVEMENT-PINN-AUDIT-001 v1.0 (2026-10-14), `docs/audit/forensic-2026-10-14/PAVEMENT_PINN_Audit_Findings_Register.md`. Source of the v3 actions and re-prioritisation: *PAVEMENT-PINN v3 Dokumen Induk* §4.10 (pp. 27–28) and the decision log §1.3 (pp. 7–8).

C.1 How the revalidation was done

1. **Change detection.** `find src server tests -newer <audit findings register>` returned no file: no source, server or test file was modified after the audit register was written (2026-10-14 23:13). `git status` shows the same 24 modified, 1 deleted and 10 untracked paths the audit recorded, plus the audit folders themselves. The audited code is therefore the current code.
2. **Spot re-verification.** Every finding whose evidence is a code location was re-checked with a read-only `grep` on 2026-10-15 (commands and hits in Appendix H, T-11). All cited lines still exist with the same content; no finding regressed or was fixed.
3. **Re-classification against v3.** A finding is `SUPERSEDED_BY_V3` when v3 replaces the requirement it was measured against (for example, "anti-spoofing" becomes the Photo Trust Gate). Superseded does not mean closed: the code still contains the defect until the v3 replacement ships, so the *next action* column is what matters.
4. **No status was upgraded to `RESOLVED_VERIFIED`.** No defect has been corrected since the audit.

Status vocabulary: `CONFIRMED_OPEN` defect still present; `PARTIALLY_RESOLVED` decision or part of the fix exists, defect still visible; `SUPERSEDED_BY_V3` original requirement replaced by a v3 decision, replacement not yet built; `BLOCKED` cannot be verified or fixed without an external resource; `RESOLVED_VERIFIED`, `REGRESSED`, `NOT_REPRODUCIBLE` — none this cycle.

C.2 Severity changes made by v3

v3 §4.10 assigns its own priorities. Under D-01 the v3 document governs, so the v3 priority is adopted below and the original severity is kept for history.

Table C.1 — Findings whose priority v3 raised

ID	Audit severity (14 Oct)	v3 priority (§4.10)	Reason implied by v3
F-06	P1	P0	Unsourced "melebihi batas ijin" verdicts are a prohibited claim (§1.5)
F-07, F-18, F-22	P1 / P1 / P2	P0	RUL in days, pothole horizon and APBN savings are prohibited claims (§1.5)
F-12, F-14	P1	P0	Unmeasured NPU/latency figures are prohibited claims (§1.5, D-11)
F-13	P1	P0 (with F-04)	Fake IR export is part of the fake training story
F-15	P1	P0	Source-of-truth conflict is the first decision (D-01, D-03)
F-16, F-19, F-20	P1	P0	"FWD Validated", agency names and random attributions are prohibited claims
F-21	P2	P0 (with F-02)	Determinism is a design principle (§4 "Prinsip 4")

C.3 Crosswalk

Table C.2 — Previous findings, current evidence and next action

ID	Original finding (abridged)	Orig. sev.	Previous evidence	Current evidence (2026-10-15)	Current status	Next action (v3 task)
F-01	Closed-form formula presented as a trained PINN	P0	`pinnEngine.ts` (296 lines, no weights); README l.1	Unchanged. README l.1 still "Physics-Informed Neural Network Decision Support System"; `assessment.ts:17` labels engine RULE_BASED honestly	CONFIRMED_OPEN	R0-01: relabel all legacy "PINN/neural" text until Module D exists; later label "PI-surrogate vX (dilatih dari MLET)" (D-02)
F-02	Fabricated telemetry (loss, latency, power, NPU)	P0	`pinnEngine.ts:203-240`, Navbar, dashboards, PDF	`pinnEngine.ts:217` latency jitter, `:238-239` random losses still present; 15 `Math.random` call sites remain in 4 runtime files	CONFIRMED_OPEN	R0-02 with `no-fake-claims.test.ts`
F-03	Strains 1–2 orders below layered-elastic theory	P0	`evidence/compare-grid.json`: engine 20/17 με vs reference 303/793 με	Engine unchanged; v3 §4.4 reports PyMastic 303/793 με for the same case, matching the audit solver (M-01 in Appendix D)	CONFIRMED_OPEN (reconfirmed by a second, independent solver)	R1-03: replace with `/v1/assess` (MLET)
F-04	Simulated WIM scraping + "PINN training" modal	P0	`BinaMargaWimPipelineModal.tsx`, `binaMargaWimScraper.ts:70-155`	7 `Math.random` calls in `binaMargaWimScraper.ts` (l.72-110)	CONFIRMED_OPEN	R0-03: delete modal and service; route returns 404
F-05	CV class/confidence from file-name keywords	P0	`cvDamageClasses.ts:24-244`	Unchanged	CONFIRMED_OPEN	R0-04: show "belum diklasifikasi" until Module C; confidence only with `modelStatus=REAL`
F-06	Strain limits 180/200/250/350 με untraceable	P1	`pinnEngine.ts:128-132`	`pinnEngine.ts:128-129` still prints "melebihi batas ijin Bina Marga (200 με)"	CONFIRMED_OPEN	R0-05 (text) then R1-05 (cited transfer function, criterion shown)
F-07	RUL/pothole horizon/PCI projection heuristic	P1	`pinnEngine.ts:90-112,173-186`	Unchanged	SUPERSEDED_BY_V3 (D-05, D-06) — defect still shipped	R1-06: remove fields; Indeks Umur P10/P50/P90 replaces them
F-08	Two standard axles (10 t vs 8.16 t)	P1	`pinnEngine.ts:54`; `esal.ts:12`	`pinnEngine.ts:54` still `axleLoadTon / 10.0`; `esal.ts:12 STANDARD_AXLE_TON = 8.16`	CONFIRMED_OPEN	R1-07: single constant 8.16 t with citation, cross-module test
F-09	Load model lacks q, a, configuration, ν	P1	`pinnEngine.ts:56-80`	Unchanged; v3 reference `mlet_strains` takes q, tyres, dual spacing, ν per layer	CONFIRMED_OPEN	R1-03 (input contract of `/v1/assess`)

ID	Original finding (abridged)	Orig. sev.	Previous evidence	Current evidence (2026-10-15)	Current status	Next action (v3 task)
F-10	Linear moisture/temperature heuristics; no pore pressure	P1	`pinnEngine.ts:35-50`	Unchanged	SUPERSEDED_BY_V3 (§1.4: pore pressure → RENCANA; wet = soaked-CBR factor)	R1-08: remove linear rain factor; wet/dry scenario with editable soaked factor labelled ASSUMED
F-11	`/api/simulate` accepts any input	P1	`simulationController.ts`	`server/routes/pinn.routes.ts:7` still registers `API_PATHS.simulate`	CONFIRMED_OPEN	R1-09: retire route or full schema validation (400/422)
F-12	Device detector asserts NPU by default	P1	`deviceDetector.ts:14-15,46`	`deviceDetector.ts:15` default "Intel® Core™ Ultra Processor with NPU..."; `:54` recommends NPU	CONFIRMED_OPEN	R0-06: browser shows "tidak diketahui"; devices only from Python `core.available_devices` (R5-02)
F-13	Hand-written IR XML with custom op	P1	`openvinoExportService.ts:15-42`	`openvinoExportService.ts:29` still `type="PhysicsPDEOp" version="custom"`	CONFIRMED_OPEN	R0-03: endpoint 404
F-14	No OpenVINO runtime/benchmark; "< 10 ms", "8.4 ms"	P1	repository-wide; `simulatedBenchmark.ts`	`simulatedBenchmark.ts` present; `demoAccounts.ts:128` (uncommitted file of another session) contains "Latensi <10 ms, NPU/CPU"	CONFIRMED_OPEN	R0-02 (remove text) and R5-01 (real `measurements/*.json`)
F-15	Blueprint vs code source-of-truth conflict	P1	`karawangPilot.ts`; `screening.ts:40,83`; `narrativeService.ts:9`	Decision now exists in v3 (D-01, D-03, D-10) but code unchanged: `screening.ts:83` marks Narogong reports OUT_OF_SCOPE; `narrativeService.ts:9` still `gemini-3.7-flash`	PARTIALLY_RESOLVED (decided, not implemented)	R0-07 (corridor config), R4-03 (cloud call off), R0-08 (DECISION_LOG merge)
F-16	No empirical data; "FWD Validated", WIM stations	P1	`LeafletMap.tsx:757,988`	`LeafletMap.tsx:26,282-287,362` still builds WIM-station correlations with licence plates	CONFIRMED_OPEN	R0-09: remove badges/stations; data inventory
F-17	Anti-spoofing absent	P1	repository-wide	Unchanged; photo checks are magic-byte, 5 MB and duplicate hint only (`photoValidation.ts`)	SUPERSEDED_BY_V3 (D-08 Trust Gate)	R3-02 (Module A with FAR/FRR)
F-18	Overlay/material from PCI look-ups	P1	`pinnEngine.ts:115-125`	Unchanged	SUPERSEDED_BY_V3 (D-14 option comparison)	R1-06 remove; R6-06 option comparison
F-19	"Official" PDF with simulated telemetry, agency names	P1	`pdfGenerator.ts:78-163`	`pdfGenerator.ts:163` "...sistem kecerdasan buatan terpadu PUPR, Kemenhub, & Intel OpenVINO"	CONFIRMED_OPEN	R0-10: status block + "bukan dokumen resmi" on every page; snapshot test
F-20	Random "PINN contribution" percentages	P1	`CommunityReportsDashboard.tsx:259-261`; `LeafletMap.tsx:360-361`	Same lines still call `Math.random`	CONFIRMED_OPEN	R0-02

ID	Original finding (abridged)	Orig. sev.	Previous evidence	Current evidence (2026-10-15)	Current status	Next action (v3 task)
F-21	Non-deterministic payloads	P2	`pinnEngine.ts:217,238-239,284`	Unchanged (`SIM-${Date.now()}` at l.284)	CONFIRMED_OPEN	R0-02 (payload equality test)
F-22	Invented economic/Zero-ODOL figures	P2	`pinnEngine.ts:188-201`	Unchanged	SUPERSEDED_BY_V3 (§1.4: APBN savings removed)	R1-06
F-23	Missing status labels on several pages	P2	page list in audit A.2	Unchanged	CONFIRMED_OPEN	R0-11: status-label component + lint-style test
F-24	Supabase migration not applied	P2	migration file; `fileCaseStore.ts`	Unchanged; credentials still absent	BLOCKED	R7-02 after D-18 choice (Supabase or one VM)
F-25	Demo weather/traffic providers	P2	`evidence.ts:56-153`	Unchanged, honestly labelled DEMO	SUPERSEDED_BY_V3 (D-12 Module E)	R6-08 (Module E from counts × VDF)
F-26	Ageing, sub-standard alert, AC-WC/PMA not implemented	P2	repository-wide	Unchanged	SUPERSEDED_BY_V3 (D-14, D-15)	R6-06, R6-07
F-27	No reference, CV or E2E tests	P2	`tests/`	Unchanged: 226 tests, none compares against an independent reference; no Playwright, no pytest	CONFIRMED_OPEN	R1-04 (MLET reference test in CI), R7-01 (Playwright)
F-28	`assessSegment` clamps out-of-range inputs	P2	`assessment.ts:120-141`	`assessment.ts:131-132` still clamps	CONFIRMED_OPEN	R1-09 (reject, never clamp silently)
F-29	2D section shows a base the engine ignores	P3	`SimulationPanel.tsx:468,495`	Unchanged	CONFIRMED_OPEN	R1-10 (section drawn from real inputs + MLET grid, D-16)
F-30	Stale README/labels/package name	P3	`README.md:1,262`; `package.json`	README l.1 unchanged; `package.json` name `react-example`	CONFIRMED_OPEN	R0-12
F-31	Unsanitised filename query parameter	P3	`openvinoController.ts` NOT VERIFIED	Static check: `openvinoController.ts:6,11` interpolate `req.query.device` into `Content-Disposition` without whitelist (not exercised over HTTP)	CONFIRMED_OPEN (static)	Removed with R0-03; otherwise whitelist

C.4 Summary

Table C.3 — Status counts

Status	Count	IDs
CONFIRMED_OPEN	22	F-01–F-06, F-08, F-09, F-11–F-14, F-16, F-19–F-21, F-23, F-27–F-31
SUPERSEDED_BY_V3	7	F-07, F-10, F-17, F-18, F-22, F-25, F-26
PARTIALLY_RESOLVED	1	F-15
BLOCKED	1	F-24
RESOLVED_VERIFIED, REGRESSED, NOT_REPRODUCIBLE	0	—

All five original P0 findings (F-01 to F-05) remain open. Under the v3 priorities, 18 findings are P0 work (§4.10), all still open.

Appendix D — Mathematical Verification Register

Document PAVEMENT-PINN-V3-BASELINE-001 · Version 1.0 · 2026-10-15.

What was and was not run. Python is not available on this machine: `python`/`python3` resolve to the Microsoft Store stub, and the only real interpreter (`C:\Users\MSI\.local\bin\python3.11.exe`) is refused by Windows Application Control ("An Application Control policy has blocked this file", os error 4551). PyMastic, NumPy, SciPy, OpenCV, PyTorch and OpenVINO therefore could not run, and **no file of the reference ZIP was executed**. Instead, every claim that could be checked without Python was checked with two kinds of independent evidence:

- the multilayer linear-elastic reference solver written for the 2026-10-14 audit (`docs/audit/forensic-2026-10-14/evidence/mlet.mjs`; Love strain function + Hankel transform; verified against Boussinesq to a relative error of about $2{\cdot}10^{-12}$, see that audit's Appendix B). It evaluates the load axis only (r = 0), so it can reproduce isolated-tyre cases but not dual-tyre superposition;
- line-by-line JavaScript ports of the pure-math functions in `pavement_core.py` (knapsack, greedy, priority score, ESAL, number checker), in `evidence/v3check2.mjs`.

Commands, Node version and raw outputs are in `evidence/evidence-run.log`. Labels: REPRODUCED independently reproduced here; CONSISTENT agrees with theory or with a hand calculation but the exact figure was not recomputed; BLOCKED needs Python/PyMastic/hardware; FAILED independent evidence contradicts the claim; INVESTIGATE needs a source the team must supply.

D.1 Governing relations

$$a = \sqrt{\frac{P_{axle}\, g \,/\, n_{tyres}}{\pi q}} \qquad \text{(EQ-1)}$$

Contact radius of one tyre (V3 SPECIFICATION §4.4; REFERENCE ZIP pavement_core.py:22-23).

$$\varepsilon\Big(kE_1, kE_2, kE_3\Big) = \frac{1}{k}\varepsilon\Big(E_1, E_2, E_3\Big) \qquad \text{(EQ-2)}$$

Elastic similarity: scaling every modulus by k scales every strain by 1/k (V3 SPECIFICATION §4.4 D2).

$$N_f = 0.0796\, \varepsilon_t^{-3.291} E_1^{-0.854}, \quad N_d = 1.365 \times 10^{-9}\, \varepsilon_v^{-4.477} \qquad \text{(EQ-3)}$$

Asphalt Institute criteria as coded (REFERENCE ZIP pavement_core.py:48-51; Huang 2004). E in psi, strain in in/in.

$$N_f = \left(\frac{6918\,(0.856V_b + 1.08)}{E_1^{0.36}\mu\varepsilon_t}\right)^5, \quad N_d = \left(\frac{9300}{\mu\varepsilon_v}\right)^7 \qquad \text{(EQ-4)}$$

Austroads (2004 form) as coded (REFERENCE ZIP pavement_core.py:52-55). Strain in microstrain, E in MPa; SF/RF omitted in code.

$$L = \frac{\ln\left(1 + Ng \,/\, E_{yr}\right)}{\ln\left(1+g\right)}, \quad N = \min\Big(N_f, N_d\Big)\Big(1 - D_{used}\Big) \qquad \text{(EQ-5)}$$

Indicative structural life index with geometric traffic growth g (Miner's rule; V3 SPECIFICATION §4.4 D3).

$$x = \left[\ln\frac{h_1}{a}, \ln\frac{h_2}{a}, \ln\frac{E_1}{E_3}, \ln\frac{E_2}{E_3}\right], \quad y = \left[\ln\frac{\varepsilon_t E_3}{q}, \ln\frac{\varepsilon_v E_3}{q}\right] \tag{EQ-6}$$

PI-surrogate features and targets (REFERENCE ZIP train_pi_surrogate.py:11-18).

$$S = 100\left(w_H H + w_S S_{str} + w_C C + w_Q Q\right), \quad C = 1 - e^{-n/2}, \quad S_{str} = \min\left(1, \max\left(0, 1 - \frac{L_{50}}{10}\right)\right) \tag{EQ-7}$$

Transparent priority score (V3 SPECIFICATION §4.6; REFERENCE ZIP pavement_core.py:84-94).

D.2 Register

Table D.1 — Mathematical and numerical verification register

ID	Item	Source claim	Independent evidence (this baseline)	Result	Status	Open issue / next action
M-01	MLET reference case: AC 100 mm/2500 MPa, base 150 mm/200 MPa, subgrade 60 MPa (CBR 6 × 10), ν 0.35/0.35/0.45, 10 t axle, 4 isolated tyres, q 0.55 MPa	v3 §4.4: PyMastic εt 303 με, εv 793 με; AGENTS: ±1 %	Audit solver: εt 303.12 με, εv 792.67 με, a = 119.12 mm (`v3check.mjs`)	Deviation 0.04 % / 0.04 %	REPRODUCED (independent solver); PyMastic run itself BLOCKED	Team must run `test_core.py` with PyMastic (R1-02) and keep the 303/793 case as a CI test (R1-04)
M-02	Same structure, 8.16 t, 4 isolated tyres	v3 §4.4: 274 / 663 με	273.74 / 662.99 με (`v3check2.mjs`)	< 0.1 %	REPRODUCED	Use as second CI fixture
M-03	Elastic similarity (all E × 3 → strains ÷ 3)	v3 §4.4, test_core.py:10-12	Ratio εt 3.0000, εv 3.0000	Exact	REPRODUCED	Keep as property test for MLET and surrogate
M-04	Dual tyres 8.16 t, spacing 330 mm, superposition at x = 0, d/2, d	v3 §4.4: 227 / 876 με	Audit solver is axis-only; off-axis superposition not available	Direction consistent with theory (dual lowers εt at the asphalt bottom, raises εv at depth)	BLOCKED	Reproduce with PyMastic; add an off-axis capability to an independent checker if a second opinion is wanted
M-05	Units: q in kPa when E in MPa (PyMastic multiplies E by 1000)	AGENTS §4, pavement_core.py:28-29	Not checkable without PyMastic source; consistent with M-01 agreement (a factor-1000 error would shift strains 1000×)	Indirectly consistent	CONSISTENT	Unit test: q = 550 kPa with E in MPa reproduces M-01
M-06	Sign convention: εt = −Strain_R (compression positive in PyMastic)	pavement_core.py:18,33	Audit solver uses tension-positive εr; εt = +303 με at the asphalt bottom	Consistent	CONSISTENT	Assert εt > 0 and εv > 0 (already in code, l.37)
M-07	Sampling depths: z = h1 − 0.01 mm (asphalt side), z = h1 + h2 + 0.01 mm (subgrade side)	pavement_core.py:25-26	Audit used ±0.001 mm with the layer index chosen explicitly; same values	Consistent	CONSISTENT	—
M-08	`iteration=20` (40 gives NaN on about 45 % of thick structures; 98.9 % valid at 20)	v3 §4.4	No PyMastic	—	BLOCKED	Record the failure rate and the failed domain in the dataset card (F-44)
M-09	Asphalt Institute fatigue and rutting constants	v3 §4.4 EQ, Huang 2004 [23]	EQ-3 coded exactly as the published form (0.0796, −3.291, −0.854; $1.365 \cdot 10^{-9}$, −4.477); MPa→psi factor 145.0377 correct	Matches	CONSISTENT	Cite edition/page in the UI and DECISION_LOG (D-05)

ID	Item	Source claim	Independent evidence (this baseline)	Result	Status	Open issue / next action
M-10	Austroads constants	v3 §4.4 "verify against the Austroads/MDP 2017 edition"	Code uses the 2004 subgrade form (9300/με)^7 and the Shell-type fatigue form without the SF/RF factor that the v3 equation shows	Spec–code mismatch on SF; edition unverified	INVESTIGATE	Team must pick and cite the edition (later editions changed the constants); add SF/RF or document its omission (F-49)
M-11	Criteria disagree by about 200× on N_d	v3 §4.4: N_d(AI) ≈ $6.7{\cdot}10^4$ vs N_d(AUS) ≈ $1.5{\cdot}10^7$ (dual 8.16 t)	From v3's dual strains: $6.66{\cdot}10^4$ vs $1.52{\cdot}10^7$ (228×); isolated 10 t case: $1.04{\cdot}10^5$ vs $3.06{\cdot}10^7$ (294×)	Order of magnitude confirmed	REPRODUCED (arithmetic)	Criterion must be chosen, cited and shown (D-05) — human decision HD-04
M-12	N_f AI vs AUS at the reference case	—	$5.40{\cdot}10^5$ vs $6.04{\cdot}10^5$ (1.12×)	Close	REPRODUCED (arithmetic)	—
M-13	Life index closed form (EQ-5)	v3 §4.4 D3	N = $5{\cdot}10^6$, E = $4{\cdot}10^5$/yr, g = 4 %: closed form 10.34 yr; explicit year-by-year sum fills 10 whole years	Correct	REPRODUCED	`mc_life.py:35` divides by ln(1+g) without the g = 0 branch that `pavement_core.life_index_years` has (F-36)
M-14	Wet subgrade (CBR 6 → 3) cuts rutting life about 4–5×	v3 §2.4, §4.4 Fig. 4.3	Isolated 8.16 t: εv 663 → 903 με, N_d(AI) ratio 3.98	Lower end of the claimed range	CONSISTENT	Depends on the ASSUMED E3 = 10·CBR and on the soaked factor 0.5 — label both
M-15	"Elastic similarity and load scaling hold exactly by construction" for the surrogate	v3 §4.4 D2 point 1	Dataset varies P (6–25 t) and q (0.5–0.8 MPa), so a ranges 76.5–197.5 mm, while the dual spacing stays 330 mm. d/a therefore ranges 1.67–4.31 and is **not** among the four features (EQ-6)	Similarity in E holds; the dual-tyre geometry is not captured, so the target is not a function of the inputs	FAILED (design defect, analytical)	Add ln(d/a) as a fifth feature or fix P and q to the standard axle used by `mc_life.py` (F-32)
M-16	Monotonicity sign table `SIGNS`	train_pi_surrogate.py:33-36; v3 §4.4 lists candidates and B.9 says enable only signs that hold on 100 % of rows	54 structures (h1 50–200 mm, E1 1–5 GPa, E2 100–300 MPa, E3 20–150 MPa), +10 % perturbation, isolated tyre, axis: εt vs E1/E3 decreases in 50/54 (**4 violations**); εt vs E2/E3 decreases 54/54; εv vs h1, h2, E1/E3 decrease 54/54; εv vs E2/E3 decreases 41/54 (**13 violations**)	2 of 6 non-zero entries fail; one of them (εt vs E1/E3) is not even in the v3 candidate list	FAILED for 2 entries	Run B.9 with PyMastic on the dual configuration before any penalty is enabled; drop failing entries (F-33)
M-17	ESAL from counts × VDF; MDP-style growth factor R	v3 §4.5 example: 1 360 normal, 2 930 actual, ratio 2.15	Port: 1 360 / 2 930 / 2.154	Matches	REPRODUCED	VDF values are illustrative, not from MDP 2017 — replace before use (Module E)
M-18	Priority score example: pothole High, AADT_norm 0.8, motorcycles 70 %, unlit, L50 1.5 yr, 3 confirmations	v3 §4.6: 83.3 (50.0 + 25.5 + 7.8)	Port: 83.3 (50.0 + 25.5 + 7.8 + 0); H before clipping = 1.022	Matches	REPRODUCED	Note: for High potholes the "unlit" multiplier saturates at H = 1; document as a policy property
M-19	Knapsack DP ≥ greedy, never over budget	v3 §4.6; test_core.py:29-37 (100 cases)	Port of the same algorithm on 1 000 independent random cases: DP ≥ greedy 1 000/1 000; strictly better 607; over budget 0; back-tracked set value equals DP optimum 1 000/1 000	Property holds	REPRODUCED (port; not the Python test itself)	Memory grows as items × budget/unit (500 items × Rp 500 bn / Rp 1 m = $2.5{\cdot}10^8$ cells): choose `unit` relative to budget (F-45)

ID	Item	Source claim	Independent evidence (this baseline)	Result	Status	Open issue / next action
M-20	Number checker accepts the good text, rejects the bad one	test_core.py:38-42	Port: good → ok; bad → rejected (950, 5, "aman")	Matches	REPRODUCED	Edge cases: a criterion name with a year ("Austroads 2004") is rejected; English "0.4" parses as 0 and is rejected; any fact value is accepted in any slot (εt value where εv belongs passes) (F-37)
M-21	Monte Carlo: deterministic seed sha256(segment:version); P10 ≤ P50 ≤ P90; Spearman sensitivity	mc_life.py	Code reading: seed deterministic for a given NumPy; percentile ordering holds by construction; wet scenario uses seed + 1, so dry/wet differ by sampling noise as well as physics; `version` is the literal "v3", not the surrogate version	Not executed	BLOCKED (execution); design issues found	Common random numbers for dry/wet; seed includes surrogate version; handle g = 0 (F-36)
M-22	Surrogate accuracy (t MAPE 1.6 %, v 0.9 %, OOD 4.8/5.4 %, 100 000 evaluations in 0.15 s)	v3 §4.4 Gambar 4.2, marked CONTOH DOKUMEN	None: no dataset, no model	—	BLOCKED	Status TARGET until the team reproduces it; must not be quoted as a team result (D-11)
M-23	PyMastic ~13 ms per case on Xeon	v3 §4.4	None	—	BLOCKED	Only from `measurements/*.json` (R5-01)
M-24	Identity of the current structural engine	audit F-01	`pinnEngine.ts` unchanged: closed-form power laws, literal coefficients, 10 t reference axle, `Math.random` telemetry	Heuristic formula; not a solver, not a trained or untrained network	REPRODUCED (static)	Replace by MLET + PI-surrogate (D-02, D-04)
M-25	Trust Gate thresholds (Laplacian variance 60 at 640 px, brightness 40–220, GPS 30 m, pHash Hamming ≤ 8)	trust_gate.py; v3 §4.1 "kalibrasi!"	Thresholds are uncalibrated initial values; out-of-corridor returns REJECT whereas v3 §4.1 expects REFERRED inside another agency's area	Not executed	INVESTIGATE	ROC on 100 sharp vs 100 blurred photos; REFERRED path (F-43)

D.3 Identity of the structural engine (D-02)

Table D.2 — What each candidate engine actually is

Engine	Where	Class	Trained?	Evidence
`runPinnSimulation`	`src/engines/pinn/core/pinnEngine.ts`	Heuristic closed-form formula	No weights exist	Audit F-01, EQ-4/EQ-5 of AUDIT-001; unchanged
`mlet_strains`	REFERENCE ZIP `pavement_core.py`	Analytical/numerical physics solver (wrapper around PyMastic)	Not applicable	Agrees with an independent solver on two cases (M-01, M-02)
`PISurrogate`	REFERENCE ZIP `train_pi_surrogate.py`	Physics-informed neural surrogate (planned)	Never trained or executed	No dataset, weights, metrics or export exist; feature set incomplete (M-15)
`life_mc`	REFERENCE ZIP `mc_life.py`	Monte Carlo wrapper over a predictor	Not applicable	Not executed (M-21)

Conclusion: the project currently has **no** trained surrogate and **no** running MLET solver. The only physics that has been numerically verified is the reference-case agreement between PyMastic (as reported by the v3 authors) and the audit's own solver. Calling anything in the current build a "PINN" or "PI-surrogate" remains a prohibited claim until R2 exit criteria are met.

D.4 Unresolved scientific questions (need a human decision or field data)

1. Which transfer-function edition governs (Asphalt Institute 1982 via Huang, Austroads 2004, or the edition adopted by MDP 2017)? The choice changes N_d by about 200× (M-10, M-11). Decision HD-04.
2. Is E3 = 10 × CBR acceptable as the default correlation, and which soaked factor applies to the Narogong subgrade? Both are ASSUMED today (M-14).
3. Should the surrogate cover variable axle loads at all? `mc_life.py` only evaluates the 8.16 t standard axle; restricting the dataset to that axle and q = 0.55 MPa would remove the d/a defect (M-15) and shrink the domain.
4. No FWD, deflection or field data exist, so every structural number is verified numerically, not validated physically. This must stay visible in the UI ("indikatif, bukan desain").

Appendix E — Consolidated Findings Register

Document PAVEMENT-PINN-V3-BASELINE-001 · Version 1.0 · 2026-10-15.

Finding IDs continue the AUDIT-001 register. F-01 to F-31 keep their original meaning (the v3 document cites them by these IDs); their current status is in Appendix C. New findings from this baseline start at F-32. Severity: P0 blocks a truthful demo or makes a false claim; P1 high; P2 medium; P3 low. "Origin" says where the defect lives: AS-IS IMPLEMENTATION (this repository), REFERENCE ZIP (candidate code from the v3 package), V3 SPECIFICATION (the document itself), or environment.

E.1 Open findings carried over from AUDIT-001 (v3 priority)

Table E.1 — Carried-over findings, ordered by v3 priority

ID	v3 prio	Title	Status	Remediation task
F-01	P0	Formula engine presented as PINN	CONFIRMED_OPEN	R0-01
F-02, F-21	P0	Random/constant telemetry; non-deterministic payloads	CONFIRMED_OPEN	R0-02
F-03, F-09	P0	Strains 15–47× too low; load model incomplete	CONFIRMED_OPEN	R1-03
F-04, F-13	P0	Fake WIM training modal; fake IR export	CONFIRMED_OPEN	R0-03
F-05	P0	Keyword "CV" with invented confidence	CONFIRMED_OPEN	R0-04
F-06	P0	Untraceable strain limits, "melebihi batas ijin"	CONFIRMED_OPEN	R0-05, R1-05
F-07, F-18, F-22	P0	RUL days, pothole horizon, PCI projection, overlay look-up, money figures	SUPERSEDED_BY_V3 (still shipped)	R1-06
F-12	P0	Browser asserts NPU	CONFIRMED_OPEN	R0-06, R5-02
F-14	P0	No benchmark; "< 10 ms"/"8.4 ms" text	CONFIRMED_OPEN	R0-02, R5-01
F-15	P0	Source-of-truth conflict	PARTIALLY_RESOLVED	R0-07, R0-08, R4-03
F-16	P0	"FWD Validated", WIM stations	CONFIRMED_OPEN	R0-09
F-19	P0	PDF with agency names, no disclaimer	CONFIRMED_OPEN	R0-10
F-20	P0	Random attribution percentages	CONFIRMED_OPEN	R0-02
F-08	P1	Two standard axles	CONFIRMED_OPEN	R1-07
F-10	P1	Linear rain factor	SUPERSEDED_BY_V3	R1-08
F-11, F-28	P1	Unvalidated `/api/simulate`; silent clamping	CONFIRMED_OPEN	R1-09
F-17	P1	No photo trust checks	SUPERSEDED_BY_V3	R3-02
F-23	P1	Missing status labels	CONFIRMED_OPEN	R0-11
F-24	P1	Persistence not applied	BLOCKED	R7-02
F-25	P1	Demo weather/traffic	SUPERSEDED_BY_V3	R6-08
F-27	P1	No reference/E2E tests	CONFIRMED_OPEN	R1-04, R7-01
F-26	P2	Early-damage flag / option comparison absent	SUPERSEDED_BY_V3	R6-06, R6-07

ID	v3 prio	Title	Status	Remediation task
F-29, F-30, F-31	P2	Section drawing, stale docs, filename parameter	CONFIRMED_OPEN	R1-10, R0-12, R0-03

E.2 New findings from this baseline

Table E.2 — New findings (F-32 to F-51)

ID	Sev.	Origin	Finding	Evidence	Impact	Remediation	Acceptance criterion
F-38	P0	Environment	No permitted Python runtime on the development machine; every Python deliverable (inference service, PyMastic, training, OpenVINO, SLM, benchmarks) is blocked here	`python`/`python3` are Store stubs; `~/.local/bin/python3.11.exe` blocked by Application Control (os error 4551)	Phases R1–R5 cannot start or be verified on this laptop; the v3 "DIUJI" status cannot be reproduced	Human decision HD-01: allow a Python 3.11/3.12 interpreter (policy exception or managed install) on the Intel Core Ultra laptop used for the interview, or name another Intel machine	`python -I -c "import numpy, scipy"` and `PYMASTIC=… python test_core.py` print `ALL TESTS PASSED` on the named machine; log stored in `evidence/03_mlet_verification/`
F-46	P1	AS-IS IMPLEMENTATION	Citizen photos are stored with EXIF/GPS metadata; no face/plate blurring; no raw-photo deletion job	`server/services/photoValidation.ts:9` ("Not done here: EXIF/GPS metadata stripping"); photo kept inside the report record (I.6)	Personal data (device, exact time, home location) retained, contrary to D-09 and UU 27/2022 minimisation	R3-01: strip EXIF server-side before persistence (until Module B ships); then Module B blur; scheduled deletion ≤ 7 days with log	Test: uploaded JPEG with GPS EXIF is persisted without any APP1/EXIF segment; deletion job log entry per photo
F-32	P1	REFERENCE ZIP	Surrogate features omit the dual-spacing ratio d/a although the dataset varies a (76.5–197.5 mm) with d fixed at 330 mm	`gen_mlet_dataset.py:11,27-28`; `train_pi_surrogate.py:11-18`	Target is not a function of the four inputs → irreducible error; the "exact by construction" claim of v3 §4.4 is false for dual tyres	Add ln(d/a) as a fifth input, or generate data only at the standard axle and q used by `mc_life.py` (HD-05)	Held-out error does not depend on P or q (stratified MAPE within ±20 % of overall)
F-33	P1	REFERENCE ZIP V3 SPECIFICATION	Monotonicity `SIGNS` contain directions contradicted by layered-elastic results; the εt-vs-E1/E3 sign is not in the v3 candidate list	`train_pi_surrogate.py:33-36`; M-16: 4/54 and 13/54 violations	Penalty would push the surrogate away from physics it is meant to respect	Run B.9 with PyMastic on the dual configuration; enable only 100 %-consistent entries; record the check	`models/sign_check.json` with per-entry counts; only 100 % entries non-zero in `SIGNS`
F-35	P1	REFERENCE ZIP	The D-04 guardrail (training-domain box, 1-in-50 cross-check against PyMastic, > 10 % → fallback) exists only in prose; no code implements `in_domain`, `crosscheck_err` or fallback	No such symbol in any reference file; `train_pi_surrogate.py` only saves `fmin/fmax`	Surrogate results could be served outside their validity domain	Implement in `/v1/assess` (R2-04)	Tests: out-of-box input → PyMastic result with `surrogate.in_domain=false`; forced 15 % disagreement → fallback flagged

ID	Sev.	Origin	Finding	Evidence	Impact	Remediation	Acceptance criterion
F-39	P1	AS-IS IMPLEMENTATION vs V3 SPECIFICATION	Workflow vocabulary conflict: current `VERIFIED` means "screening passed" (`model.ts:16-28`, spec v2.2 §24), v3 Fig. 3.2 uses `VERIFIED` for "repair verified"; v3 adds `SCREENED`, `TRIAGED`, `REJECTED_WITH_REASON`, `MERGED`, `REFERRED`	`src/domain/workflow/model.ts`, `stateMachine.ts:46-62`; v3 p. 15	Renaming or adding states touches the verified state machine, RBAC tests and the Supabase schema; a naive rename would invert the meaning of existing records	Keep current state names; add a documented display mapping now; add new states only after approval (HD-02)	Mapping table in DECISION_LOG; existing 226 tests unchanged and green; new-state tests for "no silent drop"
F-42	P1	REFERENCE ZIP	Ultralytics YOLO is AGPL-3.0; serving it from the inference service obliges source release of that service	`train_yolo.py:33`; v3 §6.5; AGENTS §9	Licence breach if integrated without a decision	Human decision HD-03: accept AGPL release, buy enterprise licence, or use a permissively licensed detector	Decision recorded before `ultralytics` is added to any requirements file
F-34	P2	REFERENCE ZIP	Surrogate training script lacks the validation that v3 §4.4 makes mandatory: no OOD split, no plain-MLP ablation, no separate validation split; `mono_penalty_positive` is a 0/1 flag, not a violation rate; `sha256_xml` hashes the FP16 `.bin`	`train_pi_surrogate.py:66,87-99`	Metrics file would not support the v3 claims	Add OOD split (train CBR ≥ 4 and h1 ≤ 180 mm; test CBR < 3 or h1 > 200 mm), ablation, violation percentage, correct hash names	`models/pi_surrogate_metrics.json` has random, OOD and ablation sections and a violation rate in %
F-36	P2	REFERENCE ZIP	`mc_life.py`: growth g = 0 divides by ln(1+0) = 0; dry and wet use different seeds (seed, seed + 1), so their difference includes sampling noise; seed "version" is the literal "v3", not the surrogate version; output lacks model version/device	`mc_life.py:13-14,27,35,41-44`	NaN/inf life for zero growth; dry→wet comparison not guaranteed monotone; provenance incomplete (AGENTS rule 3)	Reuse the g = 0 branch; common random numbers for scenarios; seed from segment id + surrogate version; add provenance fields	Tests: g = 0 finite; dry ≥ wet for every sample pair; identical inputs → identical output
F-37	P2	REFERENCE ZIP	Number checker rejects legitimate text (years in criterion names, English decimals, percentages) and accepts a fact value in the wrong slot	M-20 port results; `pavement_core.py:123-139`; `slm_brief.py:9,32`	ID/EN briefs (W7) and criterion citations will fail and fall back to the template; swapped numbers pass	Check placeholders positionally (compare filled slots to their own fact), whitelist citation years, locale-aware parsing	50-case checker suite including EN text, citations and swapped values
F-40	P2	AS-IS IMPLEMENTATION vs V3 SPECIFICATION	Decision-ID collision and superseded entries: `docs/DECISION_LOG.md` uses D-001…D-034 (e.g. D-012 separation of duties) while v3 uses D-01…D-18 (D-12 traffic load); existing D-001 ("no FastAPI yet") and D-010 ("lifecycle exactly as spec §24") contradict v3 D-04 and Fig. 3.2	`docs/DECISION_LOG.md:8,61`; v3 §1.3	Ambiguous references in commits and reviews	Import v3 decisions under a distinct prefix (`V3-D-01`) and mark D-001/D-010 superseded (proposal in Appendix G)	Every decision reference in commits resolves to exactly one entry

ID	Sev.	Origin	Finding	Evidence	Impact	Remediation	Acceptance criterion
F-43	P2	REFERENCE ZIP	`trust_gate.decide` rejects out-of-corridor photos; v3 §4.1 requires REFERRED when the point lies in another agency's area; GPS-accuracy and scene thresholds are uncalibrated	`trust_gate.py:35-44`	Valid reports for other agencies would be dropped (conflicts with "no silent drop")	Add REFERRED outcome using segment authority (D-17); calibrate thresholds by ROC	Unit tests for each decision branch; ROC report stored
F-44	P2	REFERENCE ZIP	Dataset generator silently discards failed PyMastic solves and records neither the failure rate per region nor seeds/versions beyond CPU name	`gen_mlet_dataset.py:30-31,44-47`	Training domain silently loses thick structures (v3: 1.1 % fail at iteration 20) → biased coverage and wrong domain box	Save failed rows and count; data card with seed, versions, PyMastic commit	`data/mlet_dual.card.json` lists n_requested, n_valid, failure region, versions
F-47	P2	REFERENCE ZIP	`bench_tools.py` compiles a dynamic-shape model (`[-1,4]`) on NPU and reports any exception as "not available"; NPU compilation generally needs static shapes, so a shape error would be mis-reported as missing hardware (verify on the target OpenVINO version)	`bench_tools.py:10,32-37`; `train_pi_surrogate.py:96`	Evidence page could state "NPU not available" on a laptop that has one	Reshape to the benchmark batch before compiling on NPU; record the exception text	`measurements/*_npu.json` either has latencies or an explicit error string, never a bare "not available"
F-48	P2	REFERENCE ZIP	Reference package has no licence file, no pinned dependency versions and no authorship statement for the code itself (the v3 PDF says it was drafted with Claude)	ZIP listing (`evidence/reference-inputs.txt`)	Reproducibility and GenAI disclosure (§6.6) not traceable	Add `requirements.txt` with pins, licence header, provenance note when adopting	Adopted files carry licence + provenance header; pinned lock file committed
F-49	P2	REFERENCE ZIP V3 SPECIFICATION	Austroads branch omits the SF/RF factor shown in the v3 equation and uses 2004 constants while v3 asks for the MDP 2017 edition	`pavement_core.py:52-55`; v3 §4.4 D3	Criterion "AUS" in the UI would not be the cited method	Decide edition (HD-04); implement exactly with citation	Unit test against a worked example from the cited edition
F-50	P2	REFERENCE ZIP	`train_yolo.py` states "split by location/day" and per-condition metrics but implements neither; data YAML paths are hard-coded	`train_yolo.py:1-3,21-26,36-42`	Leakage between nearby frames inflates mAP	Implement split by segment/day; per-condition evaluation; configurable paths	Split manifest committed (hashes only); metrics per class and per condition
F-41	P3	V3 SPECIFICATION	Inconsistent figures inside the v3 package: §0 says legacy strains are "14 hingga 77 kali" too small, §4.4 and AGENTS say "15 sampai 47 kali"; AGENTS Phase 0 lists F-13/F-16 as day-1 work while §4.10 merges F-13 into F-04	v3 pp. 4, 21; AGENTS §0, §5	Pitch numbers could contradict each other in front of judges	Use the reference-case ratios 15× (εt) and 47× (εv) with the case stated; record the discrepancy	One figure used consistently in deck, README and UI

ID	Sev.	Origin	Finding	Evidence	Impact	Remediation	Acceptance criterion
F-45	P3	REFERENCE ZIP	Knapsack memory grows with budget/unit × items (boolean table)	`pavement_core.py: 96-113`; M-19	Large budgets (Rp 500 bn, 500 items) need about 250 MB	Choose `unit` from budget (e.g. budget/20 000) and report rounding	Test with 500 items at the largest pilot budget completes < 2 s and < 100 MB
F-51	P3	REFERENCE ZIP	`mlet_strains` validates with `assert` (removed under `python -O`) and mutates `sys.path` at call time	`pavement_core.py: 16-19`	Silent acceptance of malformed inputs in optimised runs; import side effects	Raise `ValueError`; configure PyMastic path once at service start	Test with mismatched layer arrays raises in all modes

E.3 Risk view

Table E.3 — Count by severity (open items only)

Severity	Carried over (v3 priority)	New	Total
P0	18 (in 13 rows)	1 (F-38)	19
P1	9 (F-08, F-10, F-11, F-17, F-23, F-24, F-25, F-27, F-28)	6	15
P2	4 (F-26, F-29, F-30, F-31 — v3 groups F-29/30/31 as P2)	10	14
P3	0	3	3
Total	31	20	51

The highest risks for the next milestone are, in order: (1) false or unmeasured claims still on screen (F-01, F-02, F-05, F-06, F-14, F-16, F-19, F-20); (2) no Python runtime, so the verified-physics path cannot even start (F-38); (3) privacy of stored photos (F-46); (4) surrogate design defects that would undermine the "physics-informed" claim before a single epoch runs (F-32, F-33, F-35).

Appendix F — Development Roadmap

Document PAVEMENT-PINN-V3-BASELINE-001 · Version 1.0 · 2026-10-15. **PROPOSED TARGET STATE** — nothing in this appendix exists yet.

F.1 Principles

1. **Order follows v3 §8.1/§8.2 and AGENTS §5, adjusted to evidence.** Phase 0 (remove false claims) needs no Python and can start immediately. Phases 1–5 are blocked by F-38 (no permitted Python runtime) and must not start until HD-01 is resolved.
2. **Truthful before bigger.** A smaller system whose every number is computed or measured beats a larger one with placeholders. Optional competition items (RAG, video counting, scene classifier) stay **RENCANA**.
3. **Tracks.** Each task is tagged `SCI` (scientific correctness), `INT` (integrity/privacy/governance) or `DEMO` (competition demo value). SCI and INT work is never traded for DEMO work.
4. **Change control.** Each task is one small PR naming its finding ID (`fix(F-02): …`). Any task touching more than three source files, the state machine, RBAC, the database schema or retention policy needs a written change plan and explicit human approval first (AGENTS §6, §9). Phase 0 as a whole touches about 20 files, so it is proposed as a sequence of approved PRs, not one change.
5. **Gates.** After every phase: `npm test`, `npm run lint`, and (from Phase 1) `pytest` green; no existing test deleted or weakened; legacy tests that encode physically wrong behaviour are marked legacy with approval, never silently edited.
6. **Effort.** Sizes are relative (S ≤ 1 day, M 2–3 days, L ≥ 4 days for one developer) and assume the Python environment exists. Calendar dates are deliberately not given: the 7-day emergency plan of v3 §8.2 is not achievable as written while F-38 is open, and the interview date is not stated in the sources.

F.2 Phase table

Table F.1 — Roadmap tasks

Task ID	Source requirement	Current gap	Required change	Dependencies	Tests	Acceptance criteria	Priority · track · size
R0-01	F-01, D-02	"PINN/neural solver" text in README, legacy screens, PDF	Replace with actual status ("mesin rumus RULE_BASED, belum PI-surrogate"); roadmap wording only where labelled	none	grep test in `no-fake-claims.test.ts`	Forbidden phrases only inside roadmap-labelled copy	P0 · INT · S
R0-02	F-02, F-14, F-20, F-21, D-11	15 `Math.random` call sites; "< 10 ms"; random losses/latency/attributions	Remove generators and constants from telemetry/loss/benchmark/attribution paths; show "belum diukur"; ids/time excluded from payload equality	none	`tests/unit/no-fake-claims.test.ts`: grep for `Math.random` in runtime paths = 0; identical input → identical payload	Test green; no telemetry number on screen	P0 · INT · M
R0-03	F-04, F-13, F-31	WIM training modal + scraper; fake IR export routes	Delete modal, `binaMargaWimScraper.ts`, `openvinoExportService.ts`; routes 404	R0-02	Route test 404; module-boundary test updated with approval	Routes 404; no epoch/loss UI	P0 · INT · S
R0-04	F-05, D-07	Keyword CV with constant confidence	Show "belum diklasifikasi" until Module C; no model name/percent for citizens	none	Unit: confidence rendered only if `modelStatus='REAL'` and version	Test green	P0 · INT · S
R0-05	F-06, V3-PC-4, D-13	"melebihi batas ijin", "aman", dynamic tonnage limit + direct order text (`pinnEngine.ts:129-132,170,279`)	Remove verdict/order strings; keep numbers labelled RULE_BASED until R1	none	grep test for forbidden words	No verdict or order text	P0 · INT · S
R0-06	F-12	Browser defaults to NPU	Report "tidak diketahui" unless a device query from the inference service says otherwise	none	Unit without WebGL info → "tidak diketahui"	Test green	P0 · INT · S
R0-07	F-15, D-03	Karawang hard-coded; Narogong OUT_OF_SCOPE	`config/corridors/narogong.json`, `karawang.json`; screening reads config	Narogong geometry and authority data from team (HD-07)	Screening test: Narogong point in scope	Adding a corridor = config only	P0 · DEMO · M
R0-08	F-40, D-01	DECISION_LOG on spec v2.2; ID collision	Merge per Appendix G after approval	Approval	Doc review	Single unambiguous log	P0 · INT · S
R0-09	F-16	WIM stations with plates; "FWD Validated"	Remove; add data inventory (source, licence, status)	none	grep test	No unsupported provenance badge	P0 · INT · S

Task ID	Source requirement	Current gap	Required change	Dependencies	Tests	Acceptance criteria	Priority · track · size
R0-10	F-19, K5	PDF names agencies, no disclaimer	Status block + "bukan dokumen resmi" on every page; remove agency names	none	PDF text snapshot test	Disclaimer on each page	P0 · INT · S
R0-11	F-23	Status labels missing	One status-label component on every page showing model/engine output	none	Lint-style test over page list	All listed pages labelled	P1 · INT · M
R0-12	F-30, F-41	README title/test count, package name, inconsistent ratios	Fix docs; use 15×/47× with the case stated	none	Review	Docs match code	P2 · INT · S
R1-01	F-38	No permitted Python runtime	Provision Python 3.11/3.12 + venv on the interview laptop (policy exception or managed install)	HD-01	`python -I -c "import numpy, scipy"`	Interpreter runs under the machine policy	P0 · SCI · S (blocker)
R1-02	V3-MOD-D1	Reference never run by this team	Clone PyMastic (record commit), run `test_core.py`; store log	R1-01	`ALL TESTS PASSED`	Log in `evidence/03_mlet_verification/`	P0 · SCI · S
R1-03	F-03, F-09, D-04, V3-API-2	Heuristic engine called from 8 files (`assessment.ts`, `simulationService.ts`, 6 UI files)	`inference/` FastAPI `/v1/assess` (MLET single + dual, editable spacing); Express adapter via `INFERENCE_URL`; replace `runPinnSimulation` callers	R1-02; approval (> 3 files)	Contract tests; Express adapter test with stub; existing tests unchanged	Engineer UI shows PyMastic εt/εv with criterion and "indikatif, bukan desain"	P0 · SCI · L
R1-04	F-27	No independent reference test	pytest: 303/793 ±1 %, 274/663, ×3 similarity; Node test comparing `/v1/assess` with the audit solver on axis cases	R1-03	CI job	Fails if tolerance broken; tolerance never widened	P0 · SCI · S
R1-05	F-06, D-05, F-49	No cited transfer function	One constants module; criterion chosen and cited (HD-04); SF/RF handled	HD-04	Unit vs worked example of the cited edition	Criterion visible in UI and PDF	P0 · SCI · M
R1-06	F-07, F-18, F-22, D-05, D-06, D-14	RUL days, horizon, PCI projection, overlay look-up, money figures	Remove fields from engine, API and UI	R1-03	API test: legacy fields absent	No day-based life anywhere	P0 · SCI · M
R1-07	F-08	10 t vs 8.16 t	Single `STANDARD_AXLE_TON = 8.16` with citation used everywhere	none	Cross-module test	One constant	P1 · SCI · S
R1-08	F-10	Linear rain factor	Dry/wet scenario with editable soaked factor (ASSUMED)	R1-03	Unit: wet ≤ dry life	Labelled ASSUMED	P1 · SCI · S
R1-09	F-11, F-28	`/api/simulate` unvalidated; silent clamping	Retire route (or full schema); reject out-of-range with 400/422	approval (route removal)	Contract tests 400/422	No clamped result returned	P1 · INT · S

Task ID	Source requirement	Current gap	Required change	Dependencies	Tests	Acceptance criteria	Priority · track · size
R1-10	F-29, D-16	Section from a div	Draw from real inputs + MLET grid	R1-03	Visual review + unit on geometry	Drawing equals inputs	P2 · SCI · M
R2-01	F-32, F-44	Feature/geometry defect; silent NaN drop	Decide HD-05 (add ln(d/a) or fix axle/q); data card; failure log; generate 5 000–20 000 cases	R1-02, HD-05	Dataset card check	Card committed (hash only, data gitignored)	P0 · SCI · M
R2-02	F-33	Unchecked SIGNS	Run B.9 on the dual configuration with PyMastic	R2-01	Script output stored	Only 100 %-consistent entries enabled	P0 · SCI · S
R2-03	F-34, V3-MOD-D2	Training script incomplete; scientific core must be team-written	Team rewrites training with random, OOD, similarity, violation %, ablation	R2-01, R2-02	Metrics JSON schema test	`models/pi_surrogate_metrics.json` complete	P0 · SCI · L
R2-04	F-35, D-04	No guardrail	In-domain box + 1-in-50 cross-check + fallback inside `/v1/assess`	R2-03	OOD and forced-disagreement tests	Fallback flagged in output	P0 · SCI · M
R2-05	F-36, V3-MOD-D3	MC defects	2 000 samples, common random numbers, seed from segment + model version, g = 0 branch, Spearman	R2-04	P10 ≤ P50 ≤ P90; determinism; dry ≥ wet per sample	Tests green	P0 · SCI · M
R2-06	§5.3 point 2	No export	OpenVINO FP32/FP16 (+ INT8 via NNCF), deviation vs FP32 and vs PyMastic	R2-03	Deviation report	INT8 used only if deviation acceptable	P1 · SCI · S
R3-01	F-46, D-09	EXIF kept on stored photos	Strip EXIF before persistence (Python service, or an approved Node image library — new dependency needs approval)	approval (dependency)	Upload with GPS EXIF → stored without APP1	Test green	P1 · INT · S
R3-02	F-17, F-43, D-08, V3-MOD-A	No trust checks	`/v1/screen` gate with ROC-calibrated thresholds, REFERRED branch	R1-01, photo set	Branch unit tests; FAR/FRR report	Measured FAR/FRR stored	P0 · INT · M
R3-03	D-09, V3-MOD-B	No blur, no deletion job	Face/plate blur (licensed models), raw deletion ≤ 7 days with log	R3-01	Recall on 50 photos; deletion test	Recall reported; log entries	P0 · INT · M
R3-04	D-07, F-42, F-50, V3-MOD-C	No detector	YOLO training with split by segment/day, OpenVINO INT8, metrics per class/condition	HD-03, ≥ 100 local test photos	Metrics JSON	Confidence only from versioned model	P0 · SCI · L
R3-05	F-39, §3.3	States SCREENED/REJECTED_WITH_REASON/MERGED/REFERRED absent	Add states via approved state-machine change with mapping; "no silent drop"	HD-02 approval	Existing workflow tests unchanged + new tests	No report disappears without reason	P1 · INT · M
R4-01	D-10, V3-MOD-G	No local SLM	INT4 export; NPU → GPU → CPU fallback	R1-01, NPU laptop	Startup device test	Runs offline	P0 · DEMO · M

Task ID	Source requirement	Current gap	Required change	Dependencies	Tests	Acceptance criteria	Priority · track · size
R4-02	F-37	Checker edge cases	Positional slot check, locale, citation whitelist, 50-case suite	none	Suite	Pass rate measured	P0 · INT · S
R4-03	D-10, F-15	Cloud Gemini call active	Route narrative to `/v1/brief`; remove cloud client	R4-01, R4-02	Network-off test	No outbound LLM request	P0 · INT · S
R5-01	F-14, F-47, D-11, V3-BEN	No measurements	`bench_tools.py` fixed for static NPU shapes; run on the interview laptop	R2-06, R3-04, R4-01	Schema test of `measurements/*.json`	All §5.4 fields present	P0 · DEMO · M
R5-02	F-12	Browser guesses devices	Devices only from `core.available_devices`	R1-03	Unit	Browser never infers NPU	P0 · INT · S
R5-03	V3-API-5	No evidence page source	Evidence page reads only `/v1/measurements`; "belum diukur" otherwise	R5-01	UI test	No hard-coded numbers	P0 · DEMO · S
R6-01	O1, V3-MOD-F	No risk score	Priority score in `src/domain/intelligence/`, weights editable and audited	R2-05 (S component; 0.5 default until then)	Example 83.3; weight-change audit test	Queue ordered by score with breakdown	P0 · DEMO · M
R6-02	O2, O4, O6, D-17	No merge/SLA/referral	pHash merge, SLA per hazard, authority routing	R3-02, R3-05	18/20 merge test; SLA test; REFERRED test	Criteria of §3.4	P1 · DEMO · M
R6-03	E1, E2, V3-MOD-H	No offline field mode or correction export	Local inference + sync with version check; correction labels	R2-04, R3-04, R4-01	Wi-Fi-off E2E	Full inspection offline	P0 · DEMO · L
R6-04	W3, W4, W6, W7, V3-RAI-1	No tracking code, offline queue, confirmations, EN, safety warning	Tracking code + status page; SW queue; confirmations; ID/EN; warning before camera	R3-05	Integration + Lighthouse	Criteria of §3.4	P0/P1 · DEMO · L
R6-05	K2–K5	No optimiser/equity/disclaimed PDF	Knapsack (unit scaling F-45), scenario curve, equity map, summary + PDF	R6-01	DP ≥ greedy; curve monotone	Criteria of §3.4	P1 · DEMO · M
R6-06	E4, D-14, F-26	No option comparison	`/v1/options` with editable moduli and unit costs	R2-05	Monotonicity test	Thicker/stiffer never lower life	P1 · SCI · M
R6-07	E7, D-15, F-26	No early-damage flag	Neutral flag, ≥ 3 reports + engineer confirmation	R6-06	Word-list test	No accusatory words	P2 · INT · S
R6-08	D-12, F-25, V3-MOD-E	Demo traffic	Counts × MDP 2017 VDF (normal/actual), P10–P90	traffic survey data	Recompute-from-form test	Constants with sources	P1 · SCI · M
R6-09	K6, D-13	No advisory panel	Read-only Dishub panel	R6-08	UI review	No action buttons	P2 · DEMO · S

Task ID	Source requirement	Current gap	Required change	Dependencies	Tests	Acceptance criteria	Priority · track · size
R7-01	F-27	No E2E	Playwright golden path citizen → operator → engineer → approve → repair → verify	R6-04 (new dev dependency needs approval)	Playwright	Green in CI	P1 · INT · M
R7-02	F-24, D-18	Persistence not applied	Supabase or one VM + daily backup (HD-06)	HD-06, credentials	Integration test against chosen store	Backup restore drill	P1 · INT · M
R7-03	D-18	No public citizen URL	Deploy PWA	R7-02	Smoke test	Public URL live	P1 · DEMO · S
R7-04	V3-DOD-6, §8.3	No evidence package	`evidence/` folders of §8.3; local demo recording	all	Checklist	Folder complete	P1 · DEMO · S

F.3 Dependency chain

Table F.2 — Critical path

Step	Depends on	Why
Phase 0 (R0-01…R0-12)	approvals only	Pure TypeScript/text changes; removes every prohibited claim (V3-DOD-1)
R1-01 Python runtime	HD-01	Every Python task needs it
R1-02 → R1-03 → R1-04	R1-01	Verified physics before anything trains on it
R2-01 → R2-02 → R2-03 → R2-04 → R2-05	R1-02, HD-05	Surrogate design defects must be fixed before training (F-32, F-33)
R3-04 (detector)	HD-03 licence, local photos	Detector cannot be trained or served before the licence decision
R4-01, R5-01	interview laptop with NPU	Measurements must come from the demo machine (v3 §5.2)
R6-*	R2-05, R3-, *R4-*	Role decisions consume the new engines
R7-*	R6-*	Golden path covers the finished flow

F.4 Human approval gates

Table F.3 — Approval gates

Gate	Before	Who decides	What is approved
G-0	Any Phase 0 PR touching > 3 files	Developer + team lead	File list, tests, rollback (revert of the PR)
G-1	R1-03	Team	`inference/` service, Express adapter, `INFERENCE_URL`, retirement of `/api/simulate`
G-2	R2-03	Team	Surrogate design (HD-05), sign table, validation plan
G-3	R3-04	Team + licence owner	HD-03 Ultralytics licence path
G-4	R3-05	Team	State-machine change (HD-02), schema impact
G-5	R7-02	Team	Persistence choice (HD-06), retention policy implementation

F.5 Next milestone (recommended first phase)

Milestone M-P0 "Truthful status" = R0-01 to R0-10 merged.

Exit criteria:

1. `tests/unit/no-fake-claims.test.ts` passes: no `Math.random` in runtime telemetry/loss/benchmark/CV/attribution paths; forbidden phrases of §1.5 absent outside roadmap-labelled copy; identical inputs yield identical payloads apart from ids/timestamps.
2. `/api/openvino/export-*` and the WIM pipeline routes return 404.
3. A report located on Jalan Raya Narogong is in scope (not OUT_OF_SCOPE); Karawang still works; corridors load from `config/corridors/`.
4. Exported PDFs carry the status block and "bukan dokumen resmi" on every page and name no agency.
5. All pre-existing tests still pass (226 at the audit date plus any new ones); `npm run lint` clean.

6. DECISION_LOG merge approved and committed (Appendix G).

In parallel (no code): HD-01 (Python runtime) and HD-03/HD-04 decisions, so Phase 1 can start the day M-P0 closes.

Appendix G — Decision Log Proposal and Open Human Decisions

Document PAVEMENT-PINN-V3-BASELINE-001 · Version 1.0 · 2026-10-15.

`docs/DECISION_LOG.md` already exists (179 lines, entries D-001 to D-034, with 10 uncommitted lines from another session). It is **not modified** by this baseline. This appendix is the documented plan to update it; the merge happens only after approval (task R0-08, gate G-0).

G.1 Proposed merge

1. Add a section **"v3 decisions (from PAVEMENT-PINN v3 Dokumen Induk §1.3)"** with entries `V3-D-01` … `V3-D-18`, copied verbatim in Indonesian with the PDF page reference. The `V3-` prefix avoids the collision with the existing three-digit IDs (for example D-012 "Separation of duties" vs v3 D-12 "Beban lalu lintas") — F-40.
2. Change the header's source hierarchy to: v3 Dokumen Induk > AGENTS.md > spec v2.2 > execution brief > existing repository > engineering judgement (D-01).
3. Mark the following existing entries:

Table G.1 — Existing decisions affected by v3

Existing entry	v3 decision	Proposed marking
D-001 Keep Express; no FastAPI yet	V3-D-04 (Python inference service)	Superseded in part: Express stays the workflow API; a Python inference service is added
D-010 Lifecycle exactly as spec §24	v3 Fig. 3.2 states	Open until HD-02; keep current states until approved
D-020 Engines RULE_BASED, labelled	V3-D-02, V3-D-04	Remains true until R1-03 ships; then superseded
D-021 Traffic/WIM/weather fallbacks	V3-D-12	Superseded (Virtual WIM removed; counts × VDF)
D-022 CV stops at a surface indicator	V3-D-06, V3-D-07	Consistent; keep
D-031 Policy view shows evidence, not money	§1.4 (APBN savings removed)	Consistent; keep
D-033 Legacy illustrative screens labelled, not deleted	§1.4 HAPUS list, F-04/F-13	Superseded for the WIM modal, IR export, telemetry and attribution screens (deleted); other legacy screens stay labelled
D-012, D-013 Separation of duties; work order ≠ SPK	v3 O5	Consistent; keep (v3 builds on them)

G.2 Decisions this baseline did not take (need a human)

Table G.2 — Open human decisions

ID	Question	Options	Why it matters	Blocks
HD-01	Which Intel machine runs Python, and how is it allowed under the Application Control policy?	Policy exception for a managed Python; another team laptop; Intel Tiber AI Cloud for training only	Nothing in Phases 1–5 can run here (F-38); v3 §5.2 requires benchmarks on the interview laptop	R1-01 and everything after

ID	Question	Options	Why it matters	Blocks
HD-02	Workflow vocabulary: keep current states with a display mapping, or rename to v3 names?	(a) keep + mapping + add four new states; (b) rename with data migration	Current `VERIFIED` ≠ v3 `VERIFIED` (F-39); state machine and RBAC are the verified backbone	R3-05
HD-03	YOLO licence path	AGPL-3.0 open-source release of the inference service; Ultralytics enterprise licence; permissive detector	Legal exposure when served over a network (F-42)	R3-04
HD-04	Transfer-function criterion and edition	Asphalt Institute (Huang 2004); Austroads 2004 form; edition adopted by MDP 2017	N_d differs about 200× (M-11); must be cited and shown	R1-05
HD-05	Surrogate input domain	Add ln(d/a) as a fifth feature; or restrict data to the 8.16 t standard axle and q = 0.55 MPa	Current design cannot represent dual-tyre geometry (F-32)	R2-01
HD-06	Persistence for the pilot (D-18 leaves the choice)	Apply Supabase migration; one VM with file store + daily backup	Schema, backups, identity for internal roles (F-24)	R7-02
HD-07	Narogong corridor data	Geometry, managing authority per segment (SK penetapan status jalan), layer data with sources	D-03, D-17 need real configuration; no such data in the repository	R0-07, R6-02
HD-08	Commit attribution convention	AGENTS §6 asks for an `Assisted-by: <tool>` trailer to make GenAI disclosure countable; this repository's owner has asked that co-author trailers name Pavement-PINN	Honest GenAI disclosure (v3 §6.6) depends on a consistent, truthful convention	All commits from Phase 0
HD-09	May the reference ZIP code be adopted, and who rewrites the scientific core?	Adopt with provenance header; team rewrites core modules (AGENTS §6)	No licence/authorship in the ZIP (F-48); judges ask who wrote the science	R1-03, R2-03

Appendix H — Test and Evidence Register

Table H.1 — Commands and checks

Test ID	Command or procedure	Environment	Result	Evidence	Reproduced?
T-01	`git rev-parse --show-toplevel`, `git branch --show-current`, `git rev-parse HEAD`, `git status --porcelain --ignored`, `git log --oneline -12`, `git stash list`	repo root	HEAD `4a9b4fc`; 24 M, 1 D, 10 untracked + audit folders; ignored: `dist/`, `node_modules/`, AUDIT-001 log; no stash	this report, Document Control	Read-only
T-02	`netstat -ano` for ports 3000/8100	host	3000 LISTENING (PID 18032, user's server, untouched); 8100 free	—	—
T-03	Tool inventory: `command -v python python3 py pip uv node pdftotext pdftoppm unzip` + `uv python list`	host	Node, pdftotext, unzip, Chrome present; no pdftoppm; Python stubs only; `python3.11.exe` blocked by Application Control (os error 4551)	console	Yes (deterministic)
T-04	`sha256sum` of inputs; `unzip -lv`	inputs	11 entries, safe paths	`evidence/reference-inputs.txt`	Yes
T-05	`pdftotext -layout` and `-raw` on the v3 PDF	scratch copy	65 pages, 3 596 lines	—	Yes
T-06	`node evidence/v3check.mjs`	Node v24.16.0	Reference 303.12/792.67 με; similarity 3.0000; AI/AUS N_f, N_d; life closed form; sign check 54 structures	`evidence/evidence-run.log`	Yes (deterministic)
T-07	`node evidence/v3check2.mjs`	Node v24.16.0	8.16 t 273.74/662.99 με; CBR 6→3 N_d ×3.98; knapsack 1 000 cases; priority 83.3; ESAL 1 360/2 930; checker cases	`evidence/evidence-run.log`	Yes (deterministic)
T-08	AUDIT-001 T-03 `npm test` (cited)	same working tree, 2026-10-14	226/226 pass	AUDIT-001 Appendix C	Not re-run (no change, §1.4)
T-09	AUDIT-001 T-04 `npm run lint` (cited)	same	exit 0	AUDIT-001 Appendix C	Not re-run
T-10	Python reference suite `test_core.py`	—	Not run	—	**BLOCKED** (F-38)
T-11	`find src server tests -newer <AUDIT-001 register>`; spot `grep` of 20 finding locations; prohibited-phrase sweep	repo	No newer file; all cited lines present; 6/8 prohibited claims present	Appendix C	Yes
T-12	`npm run lint` after adding this folder (its `.mjs` files fall under `tsc` because of `allowJs`)	repo	See PDF QA summary	`PAVEMENT_PINN_v3_PDF_QA_Summary.md`	Yes

Table H.2 — Not performed, with reason

Item	Reason	Status
PyMastic, NumPy, OpenCV, PyTorch, OpenVINO, Ultralytics runs	No permitted Python runtime	BLOCKED
Dual-tyre MLET reproduction	Independent solver is axis-only	BLOCKED
Surrogate training, YOLO training, SLM, benchmarks	No Python, no data, no NPU access verified	BLOCKED
`npm run build`, deploy, migrations	Would change `dist/` or external state; not needed	Not run (safety)
HTTP calls to the user's server	Port 3000 belongs to the user	Not run
Playwright golden path	No suite exists	RENCANA

Appendix I — API and Data-Contract Differences

Table I.1 — Current Express endpoints vs v3 inference contracts

v3 contract	Closest current endpoint	Difference	Action
`POST /v1/screen` (multipart image, lat, lon, gps_acc, captured_at)	Report submission in `workflow.routes.ts` (photo as URL or base64 data URL)	No trust, privacy or detection outputs; photo inline in the case record	New service; Express forwards photo, stores blurred result only
`POST /v1/assess` (layers with h, E, ν, source; subgrade CBR + state; traffic P10–P90 + growth; criterion; n_mc; seed)	Intelligence assessment/What-If routes (`intelligence.routes.ts`); legacy `POST /api/simulate`	Current inputs: axle t, rain, flood, temperature, CBR, AC thickness/modulus; no q, a, ν, base layer, criterion, MC; outputs include legacy life fields	New service; Express adapter; retire `/api/simulate`
`POST /v1/options`	none	—	New
`POST /v1/brief` (role, facts) → text, checker, model, device, tokens/s	Narrative endpoint (Gemini, fallback)	Free-text prompt to cloud; no checker	Replace
`GET /v1/measurements`	`/api/openvino/export-*` (fake), simulated benchmark	Fabricated values	Remove, then new

Table I.2 — Data model differences (v3 §4.9 vs current)

v3 table	Current	Note
corridors, segments (authority, layers with source)	`karawangPilot.ts`, `indonesianRoads.ts` constants; migration tables (unapplied)	Needs config + provenance per layer (HD-07)
reports (tracking code, salted device hash)	`report_cases` (file store; migration unapplied)	No tracking code; account-id ownership exists
photos (blurred URL, raw URL null after deletion, pHash)	photo inside the case record	Separate storage, deletion
trust_checks, detections, assessments, confirmations, priority_snapshots, traffic_counts, measurements	`pinn_predictions` (migration), What-If runs persisted	New tables or JSON collections after HD-06
audit_log (+ weight changes, model promotion, data deletion)	audit trail exists	Extend event types

Appendix J — Glossary and Sources

Table J.1 — Glossary

Term	Meaning
MLET	Multilayer linear-elastic theory (Burmister); stresses and strains in a layered pavement under a circular load
εt / εv	Horizontal tensile strain at the bottom of the asphalt / vertical compressive strain at the top of the subgrade (με = 10^{-6})
PI-surrogate	Neural network imitating MLET, with physics entered through dimensionless variables and derivative penalties; not a PDE-residual PINN
PyMastic	Open-source Python layered-elastic solver (Apache-2.0)
Transfer function	Empirical law converting strain to allowable load repetitions (fatigue N_f, rutting N_d)
ESAL / VDF / CESA	Equivalent 8.16 t standard axle load / vehicle damage factor / cumulative ESAL
Indeks Umur	Indicative structural life index in years (P10/P50/P90), not a design life
Trust Gate	Layered photo checks with measured false-accept/false-reject rates
FAR / FRR	False accept / false reject rate
HD-xx	Open human decision (Appendix G)
R-x-yy	Roadmap task (Appendix F)
M-xx	Mathematical verification item (Appendix D)

Table J.2 — Sources

No.	Source
[S1]	*PAVEMENT-PINN v3 Dokumen Induk*, v3.0, 15 Oct 2026, 65 pp. (sha256 `1cdc4229fb…b6ab`)
[S2]	`AGENTS.md` — Panduan Agent Coding untuk PAVEMENT-PINN v3 (sha256 `06a7635c…4b535`)
[S3]	`pavement_pinn_v3_code.zip` (sha256 `664c6a7f…2b88`)
[S4]	PAVEMENT-PINN-AUDIT-001 v1.0, 2026-10-14, `docs/audit/forensic-2026-10-14/`
[S5]	Huang, Y. H., *Pavement Analysis and Design*, 2nd ed., Pearson, 2004 (Asphalt Institute criteria), as cited in [S1] ref. [23]
[S6]	Austroads, *Guide to Pavement Technology Part 2*; Bina Marga, *Manual Desain Perkerasan Jalan 2017*, as cited in [S1] ref. [24] — edition to be fixed by HD-04
[S7]	PyMastic, github.com/Mostafa-Nakhaei/PyMastic, as cited in [S1] ref. [18]
[S8]	Repository `pavement-pinn`, commit `4a9b4fc`, working tree of 2026-10-15